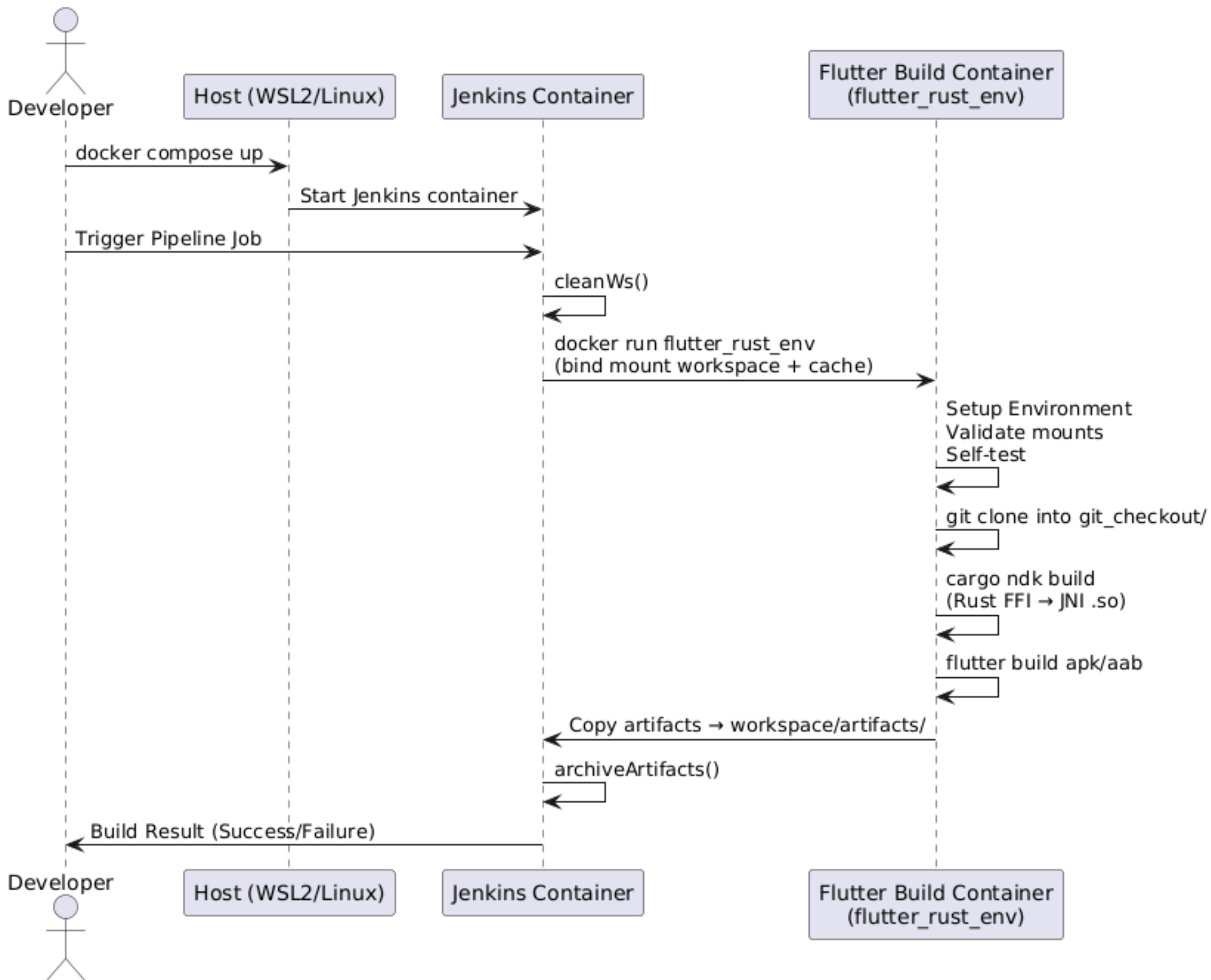
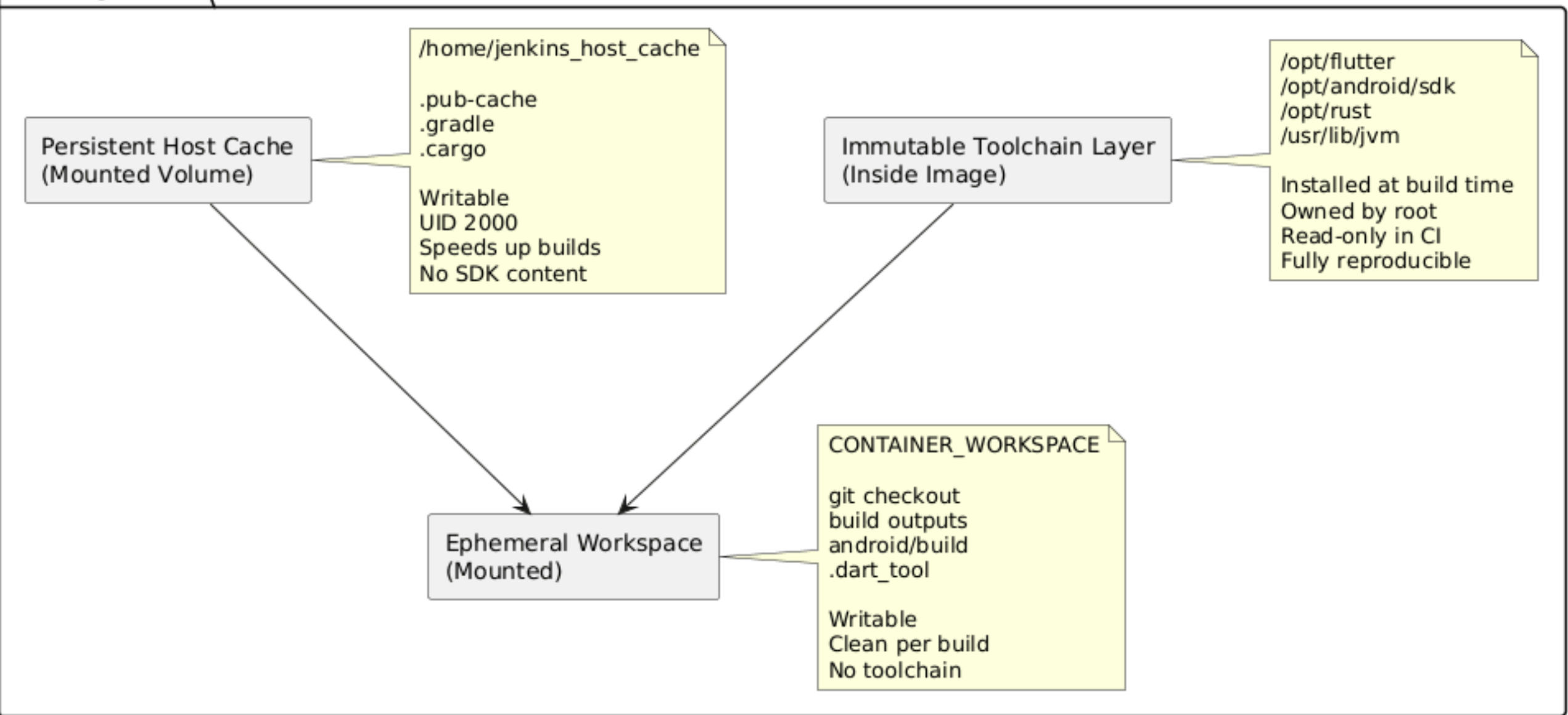


Deployment Sequence - Host → Jenkins → Flutter Container

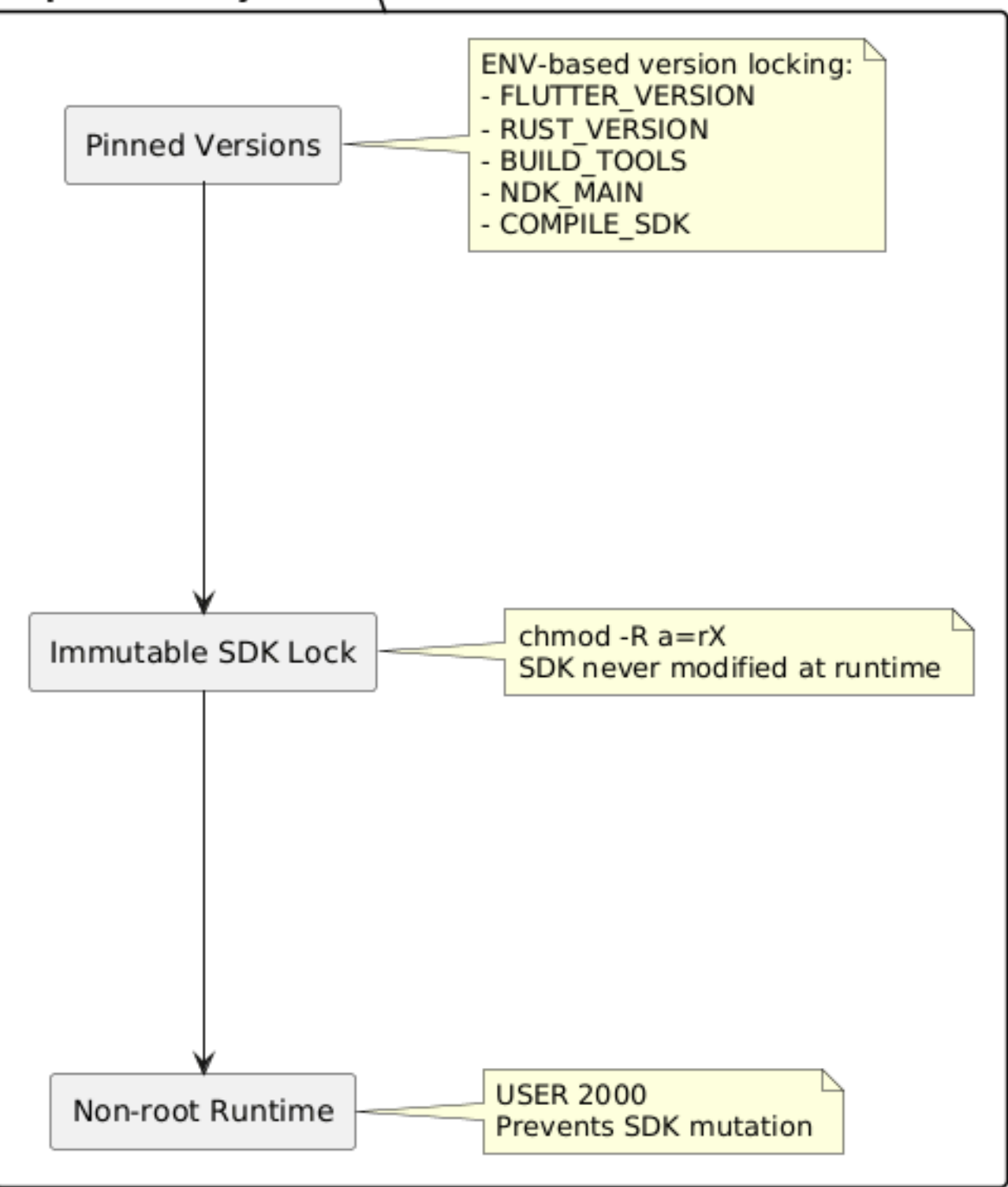


Immutable Toolchain Versions	
Java	: 17
Flutter	: 3.35.7
Rust	: 1.91.1
Android SDK Tools	: 13114758
Android Compile	: 36
Android BuildTools	: 35.0.0
NDK	: 28.2.13676358
CMake	: 3.22.1
Base Image	: ubuntu 22.04

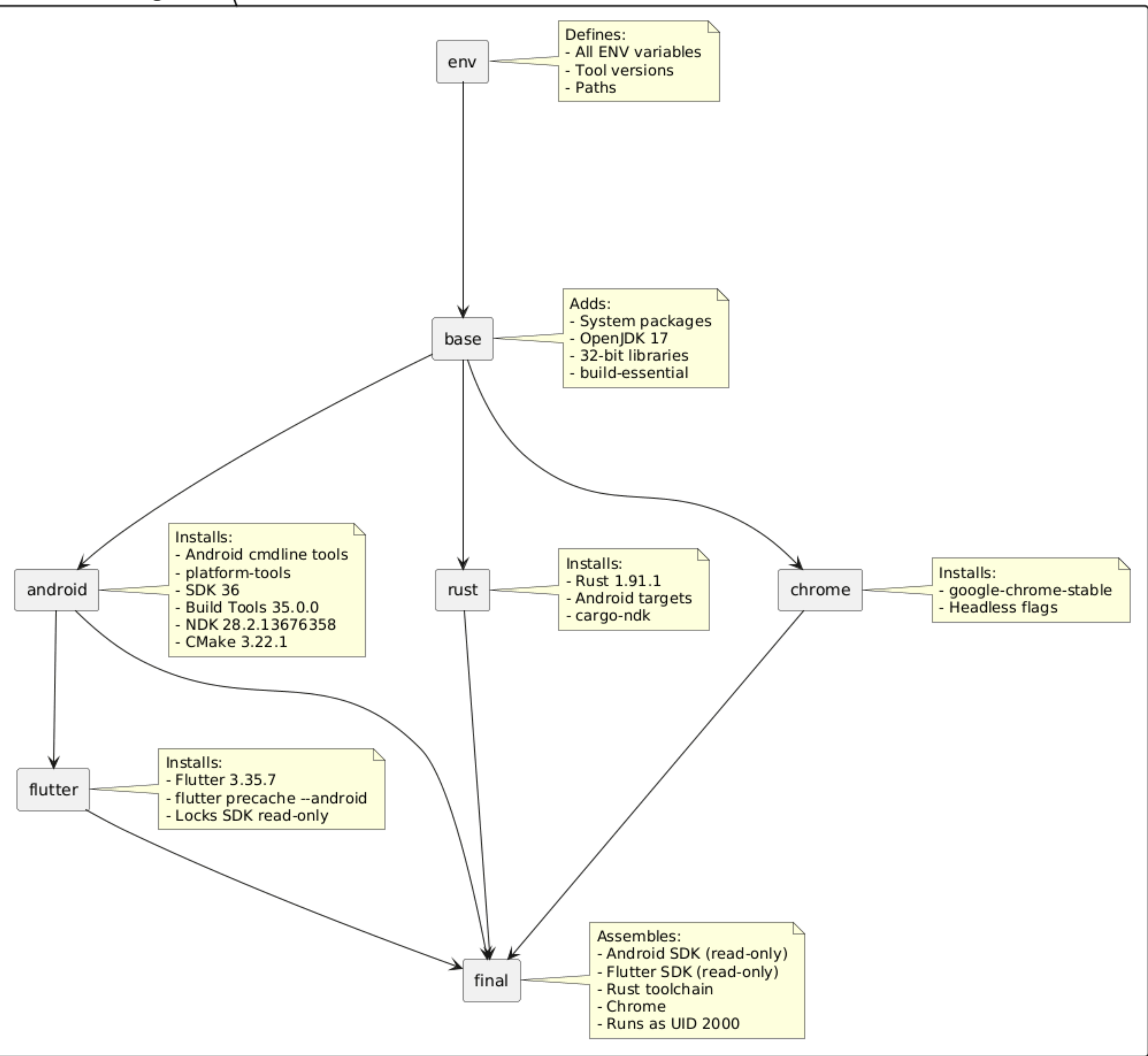
Security Model



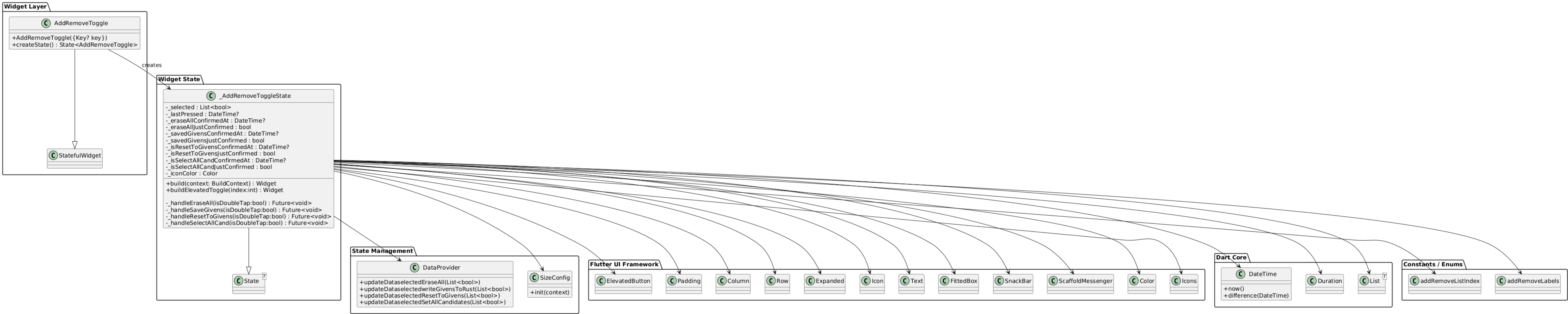
Reproducibility Model



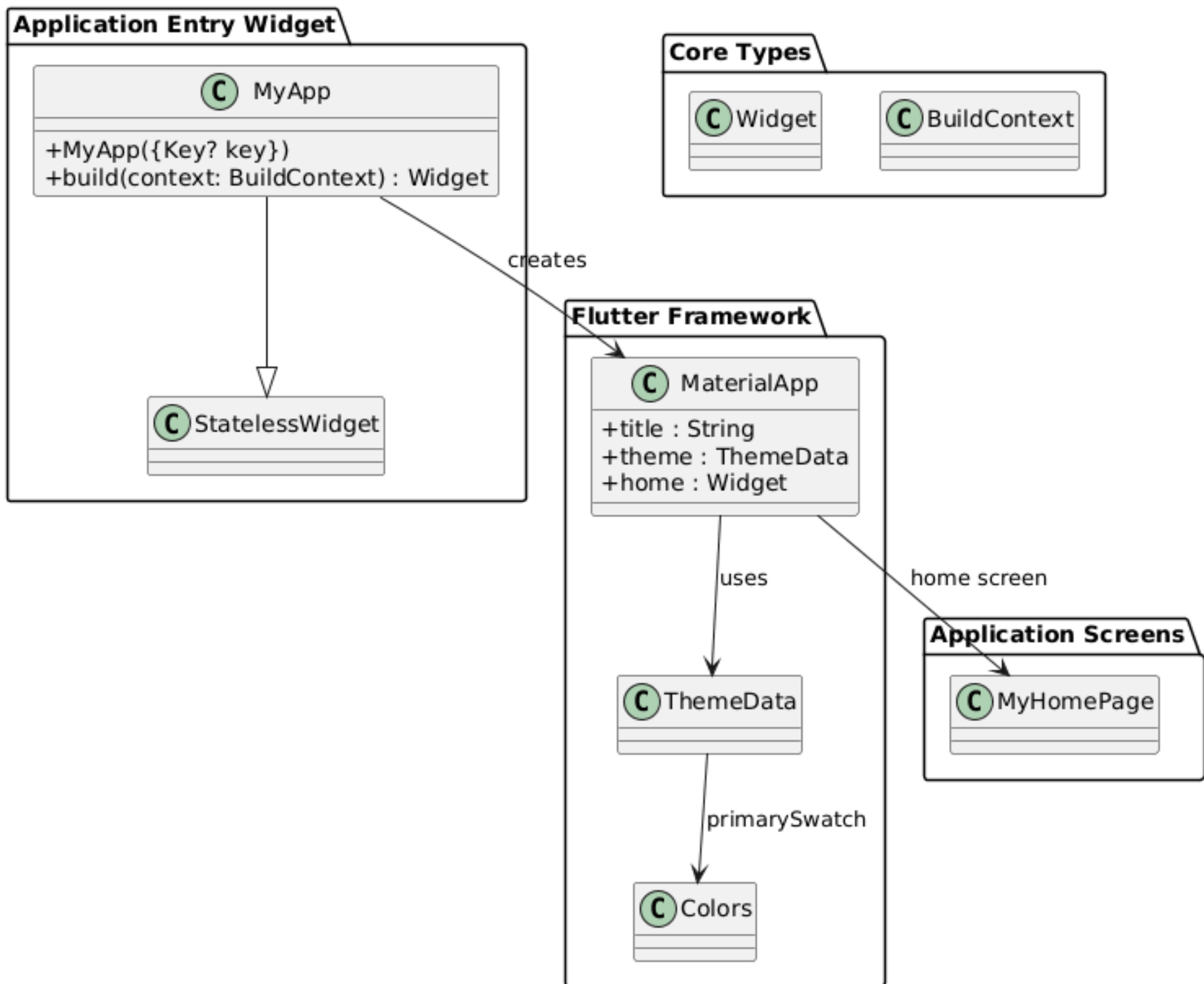
Docker Multi-Stage Build



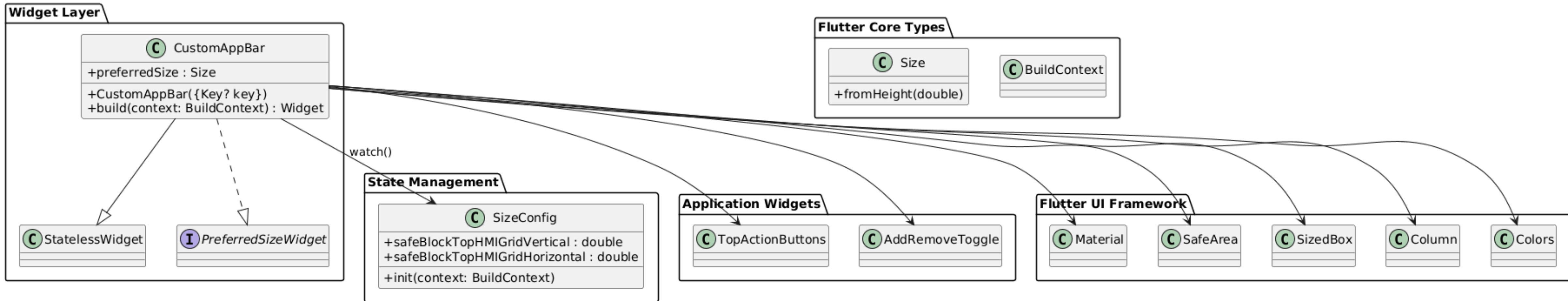
Flutter Widget Class Diagram - AddRemoveToggle



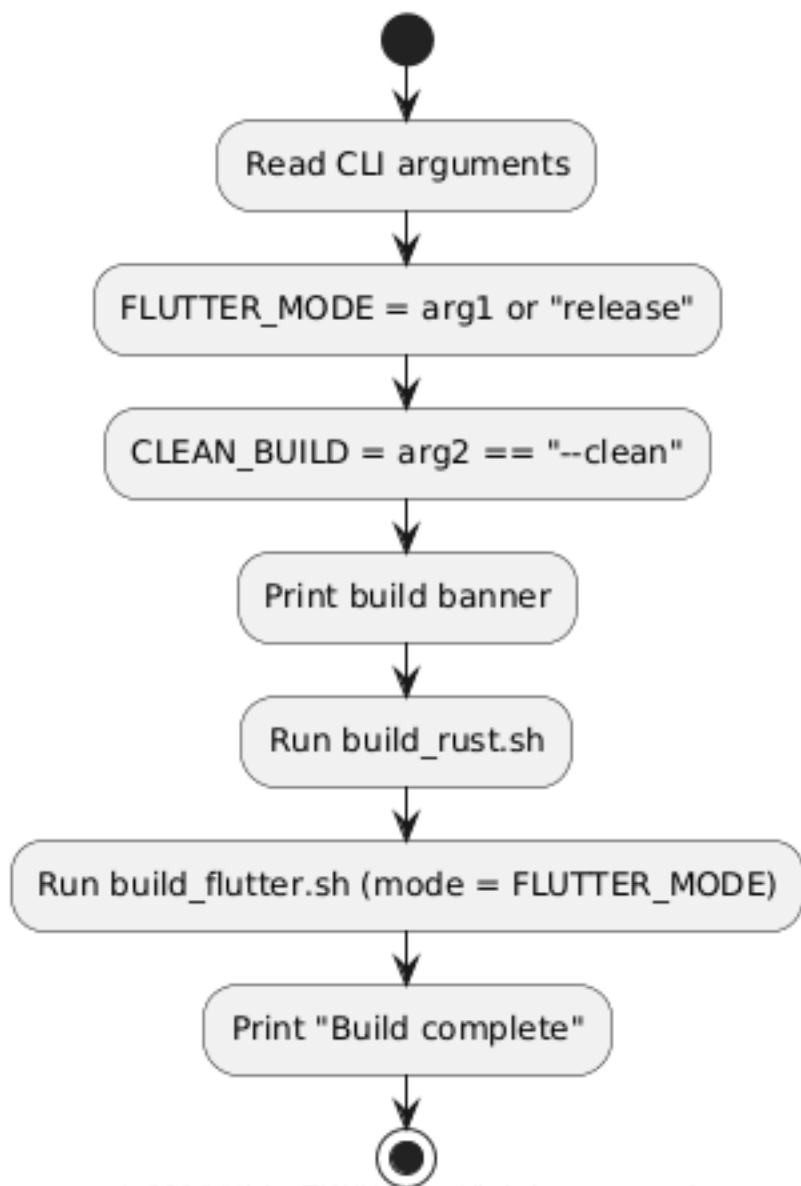
Flutter Widget Class Diagram - MyApp



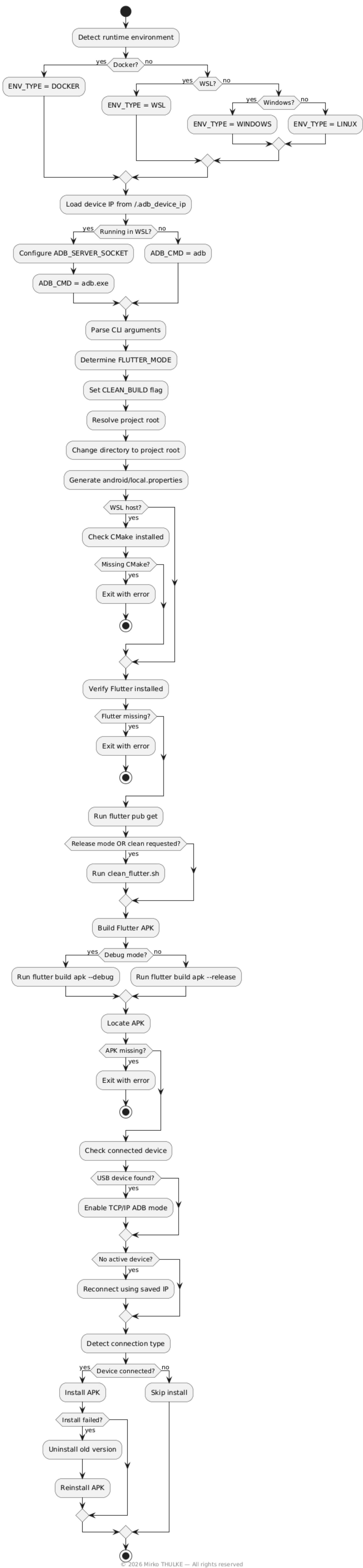
Flutter Widget Class Diagram - CustomAppBar



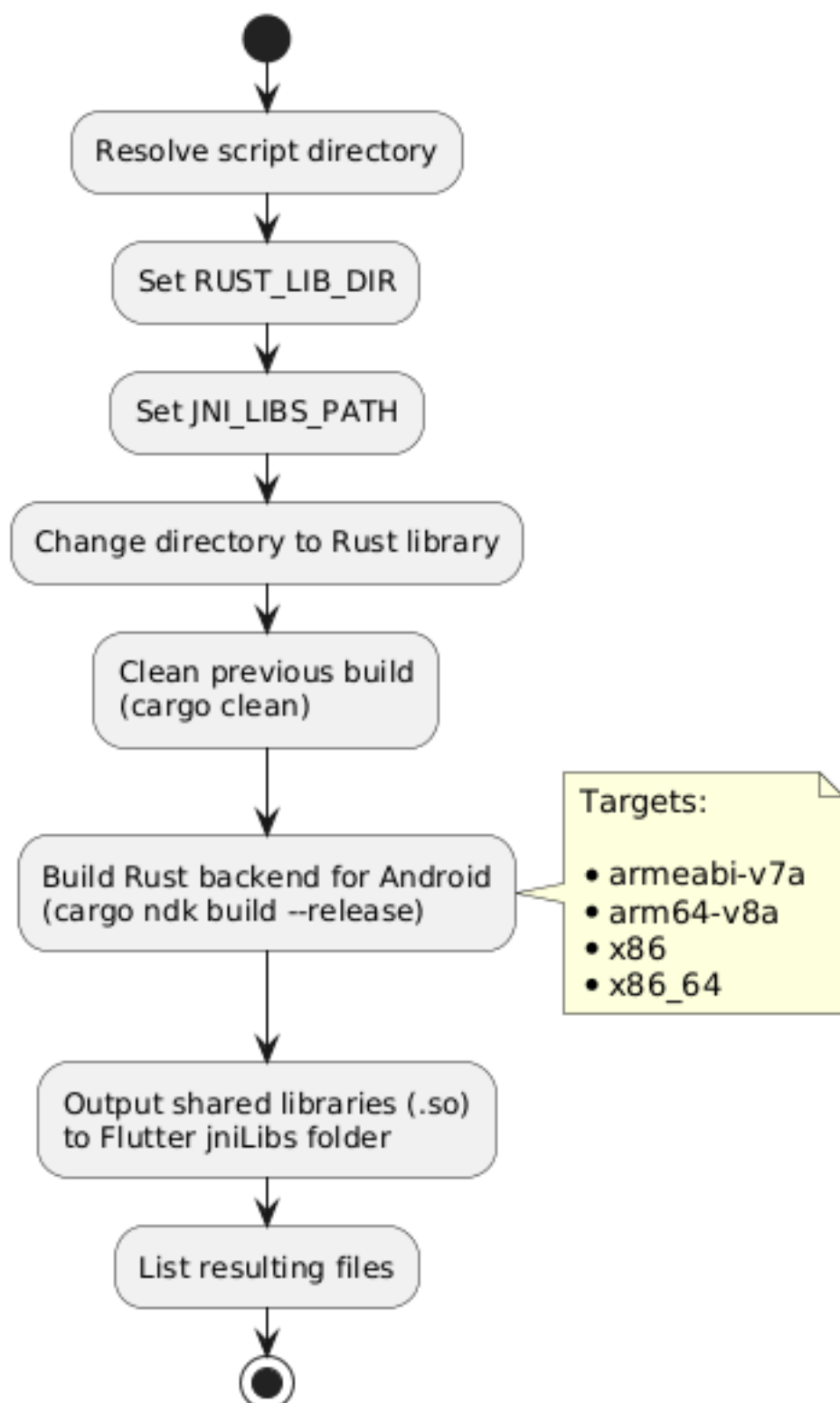
Rust + Flutter Build Pipeline



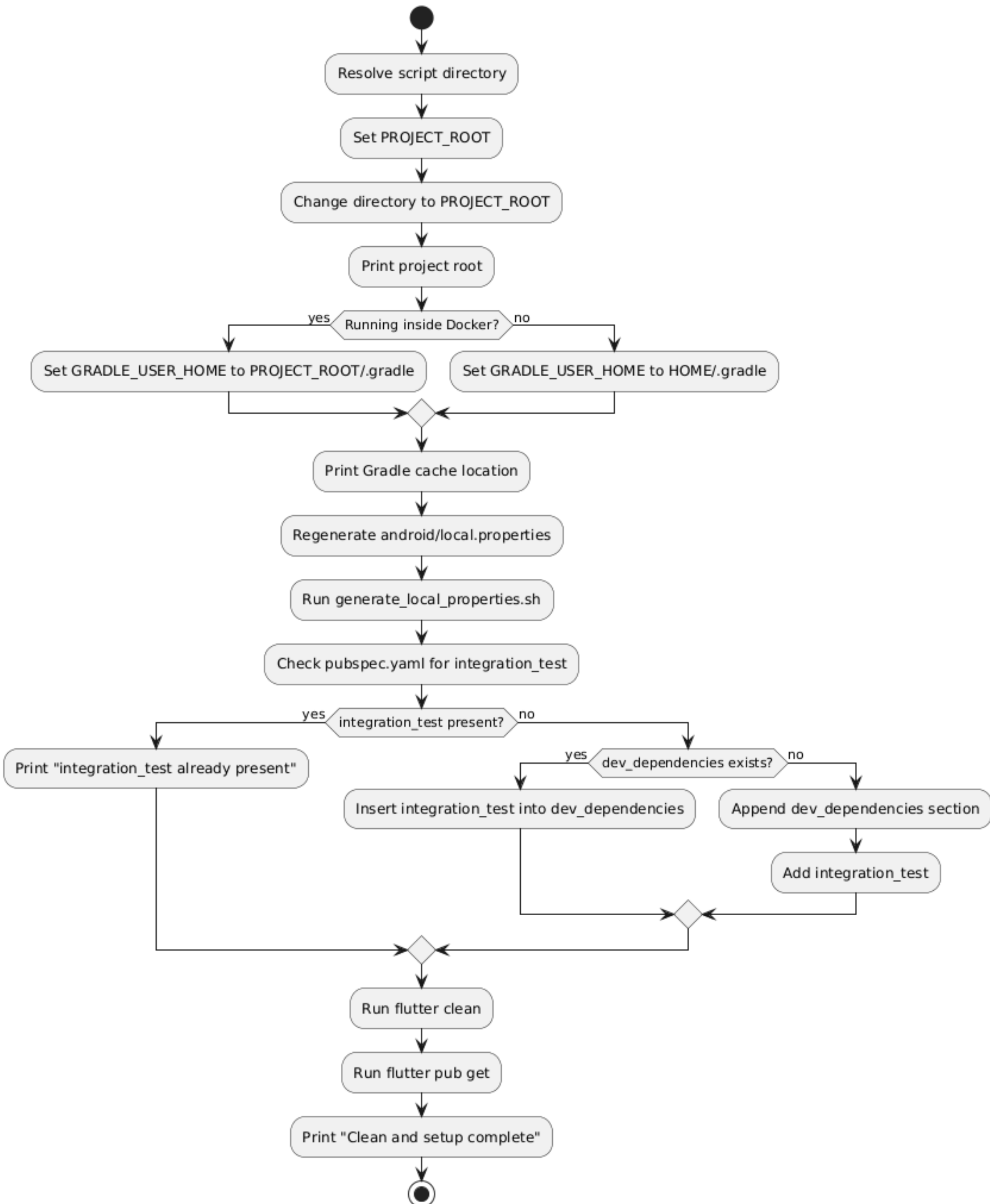
Flutter + Rust Build & Deploy Script



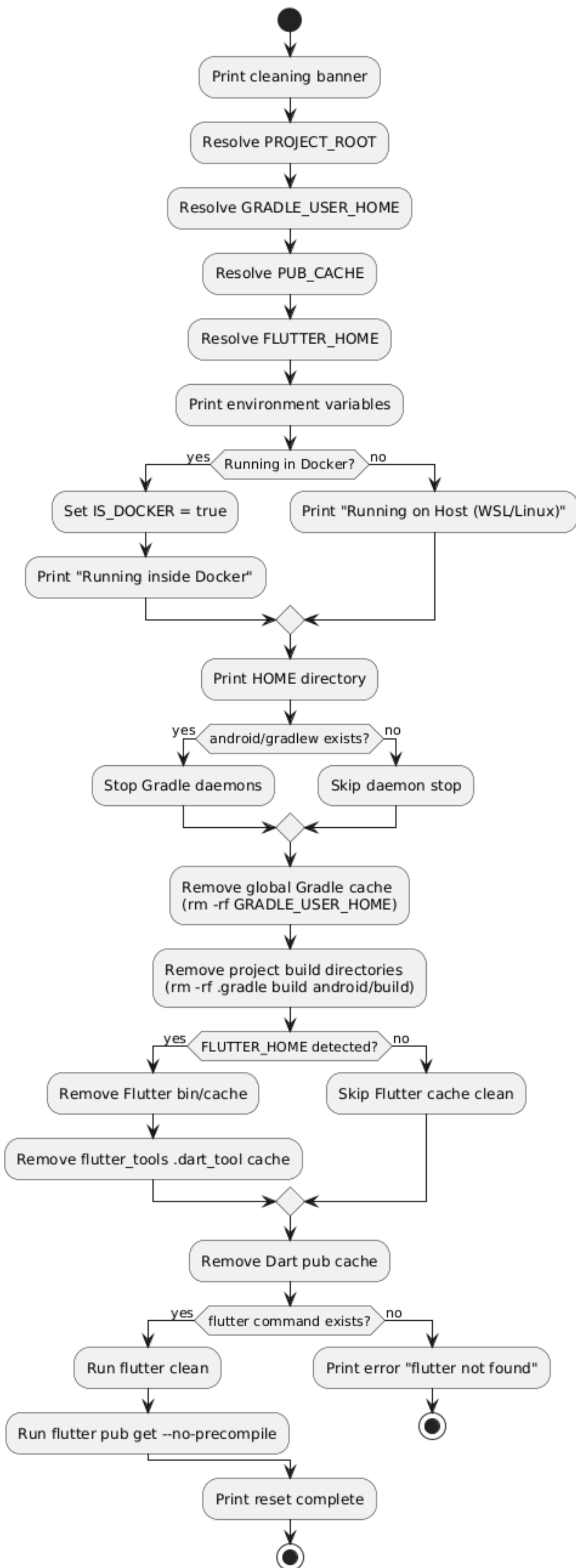
Rust Android Build Pipeline



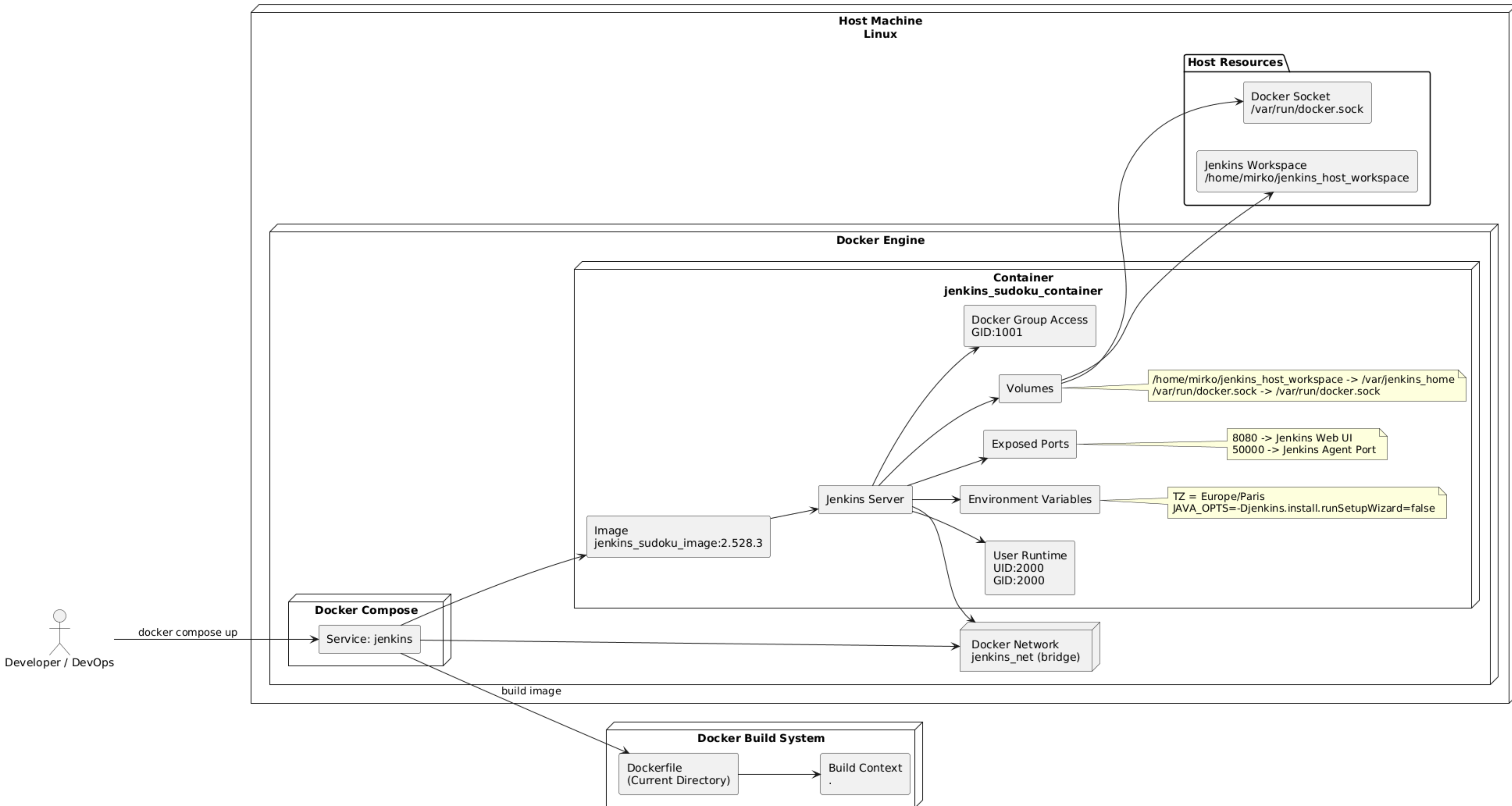
Flutter Project Setup Script



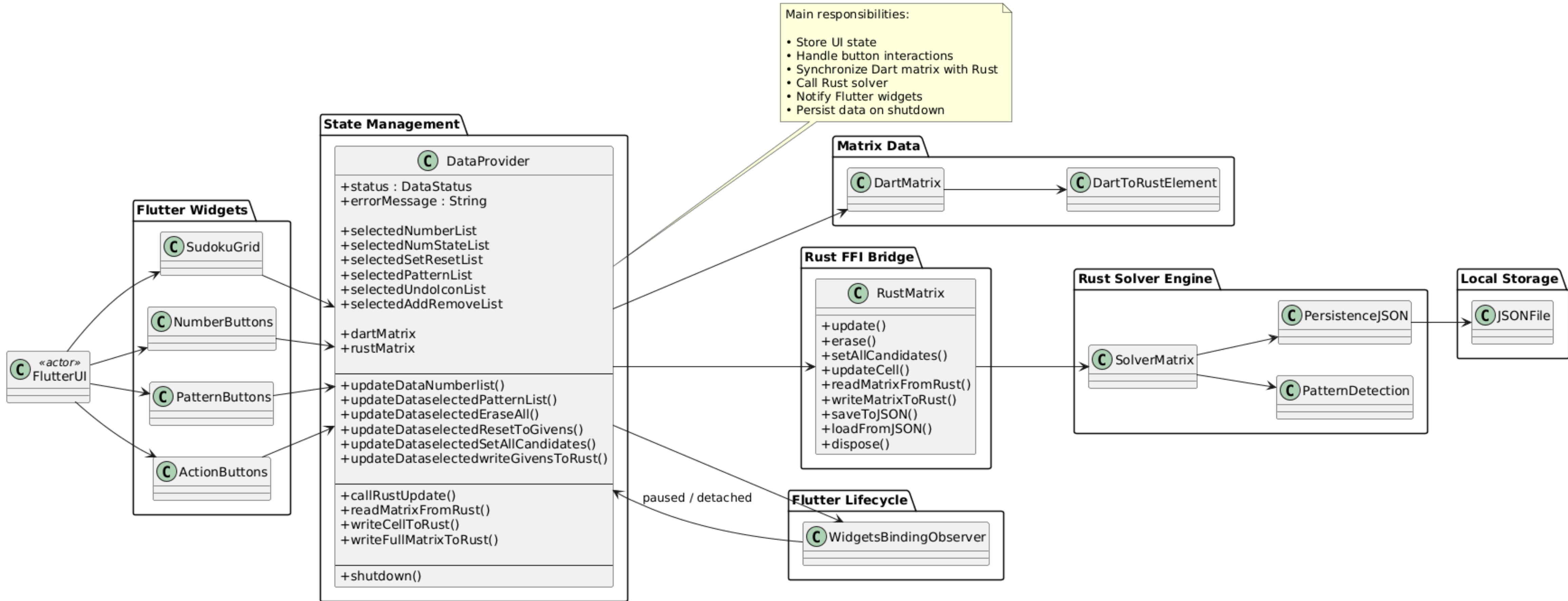
Flutter & Gradle Cache Reset Script



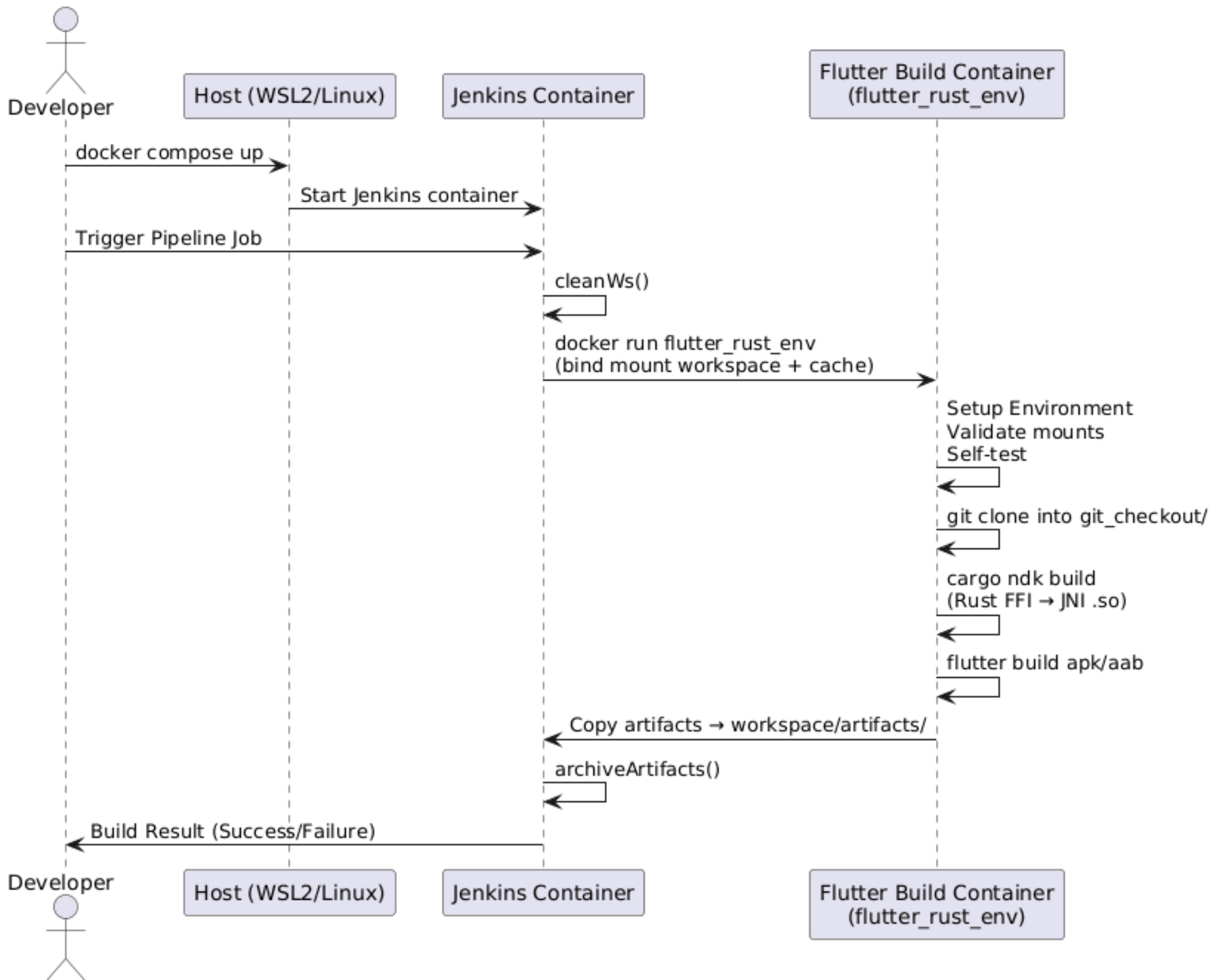
Jenkins Sudoku CI - Docker Compose Deployment



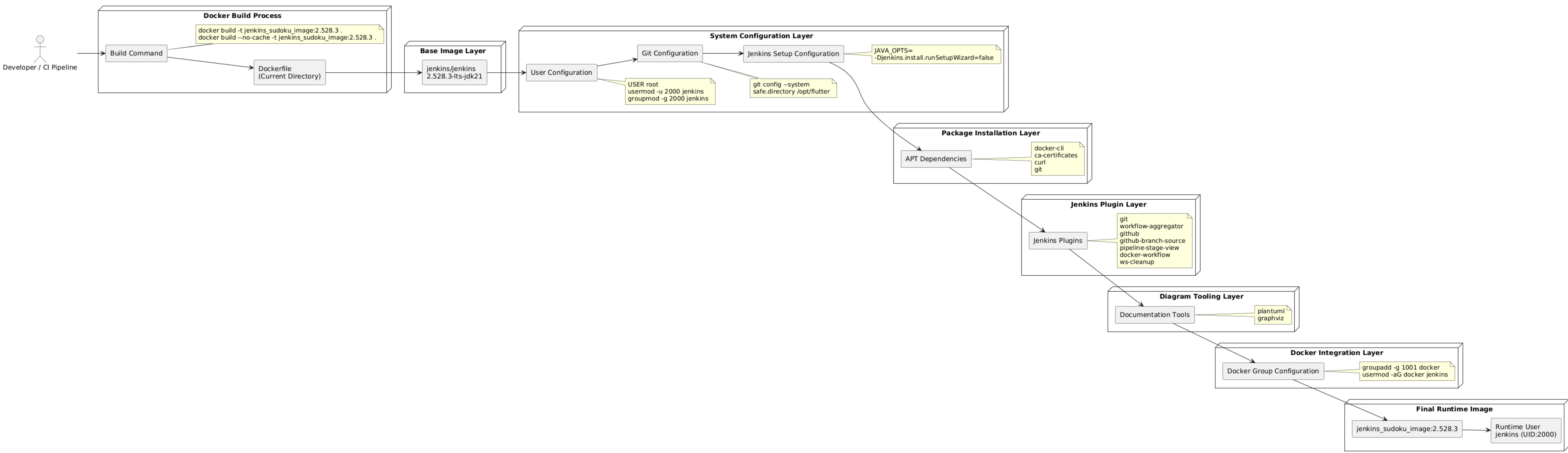
Flutter DataProvider - State Management & Rust FFI Bridge



Deployment Sequence - Host → Jenkins → Flutter Container

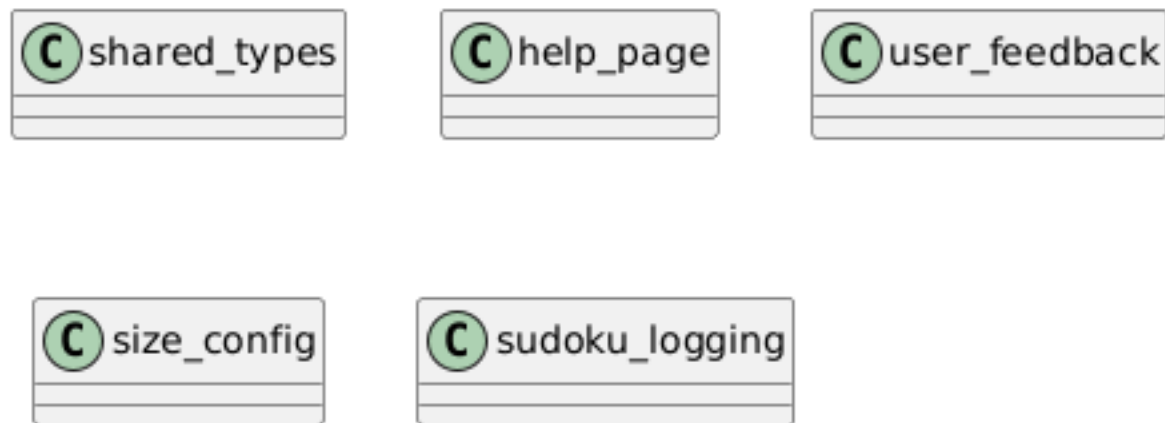


Jenkins Sudoku CI - Docker Image Build

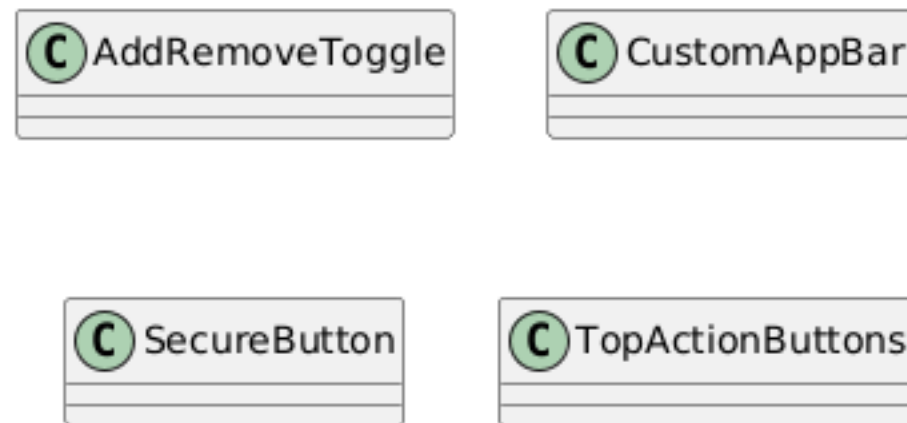


Dart Export Module

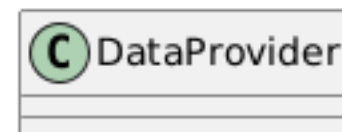
utils



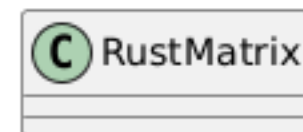
home



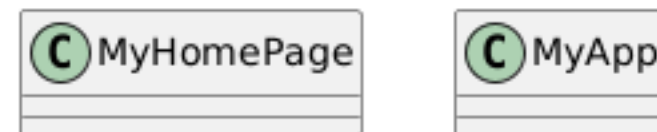
providers



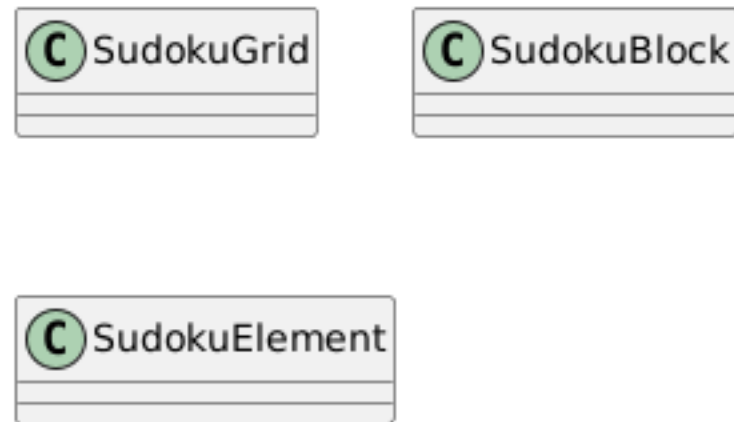
rust_ffi



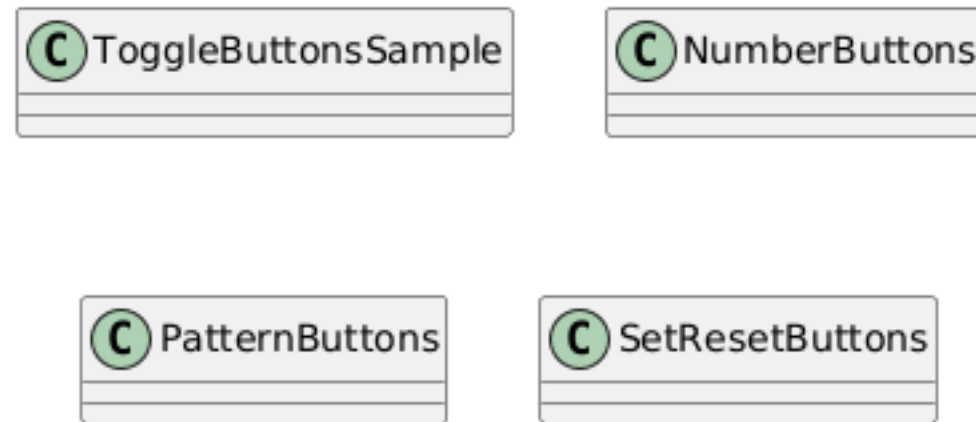
app



sudoku

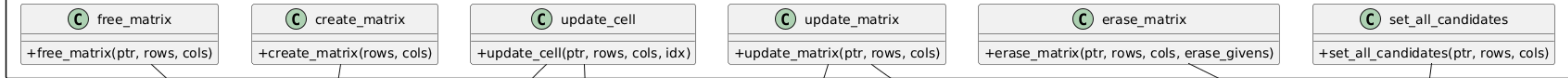


sudoku_buttons

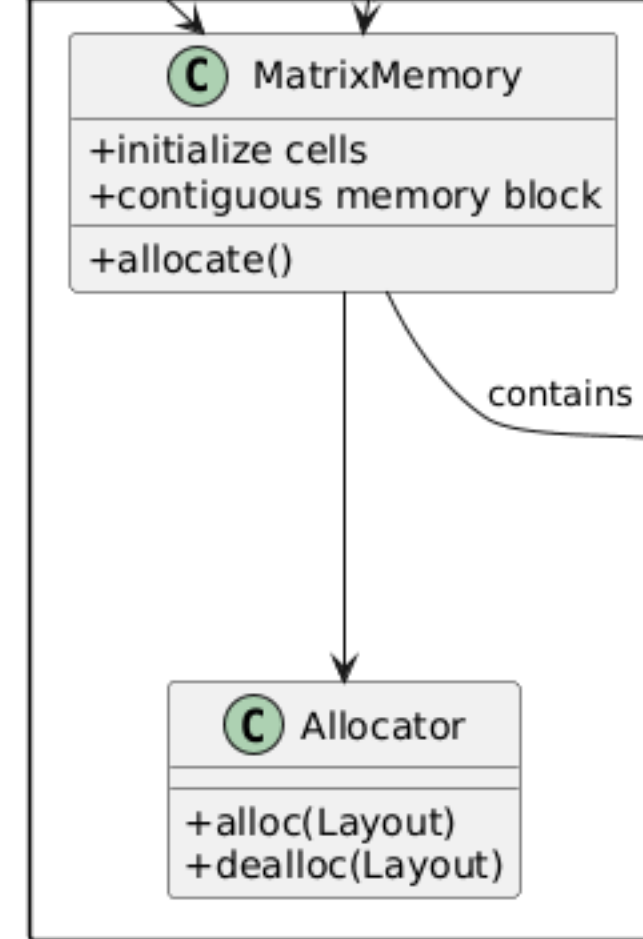


Rust Sudoku Matrix FFI Architecture

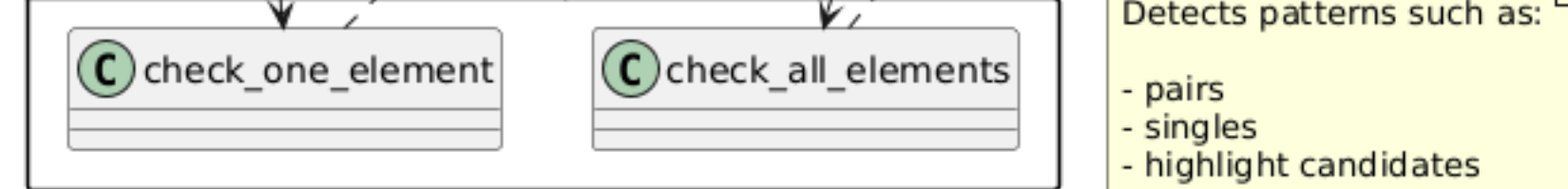
C FFI Interface



Memory Model



Pattern Detection Engine

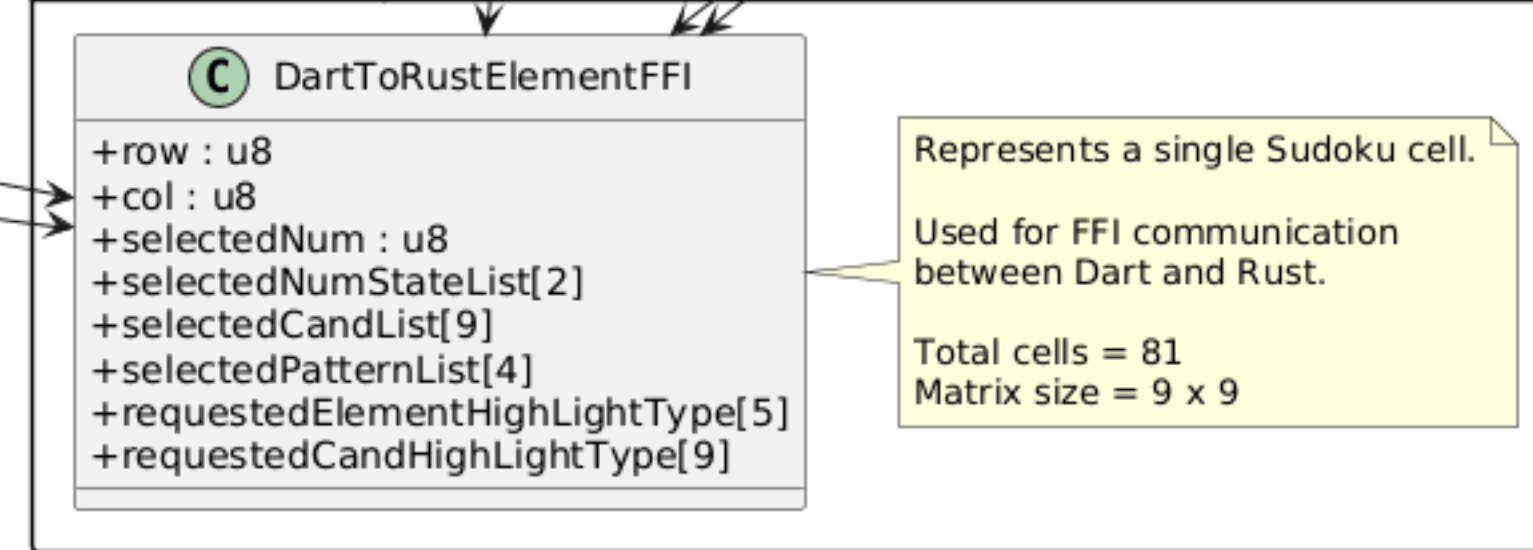


Detects patterns such as:

- pairs
- singles
- highlight candidates

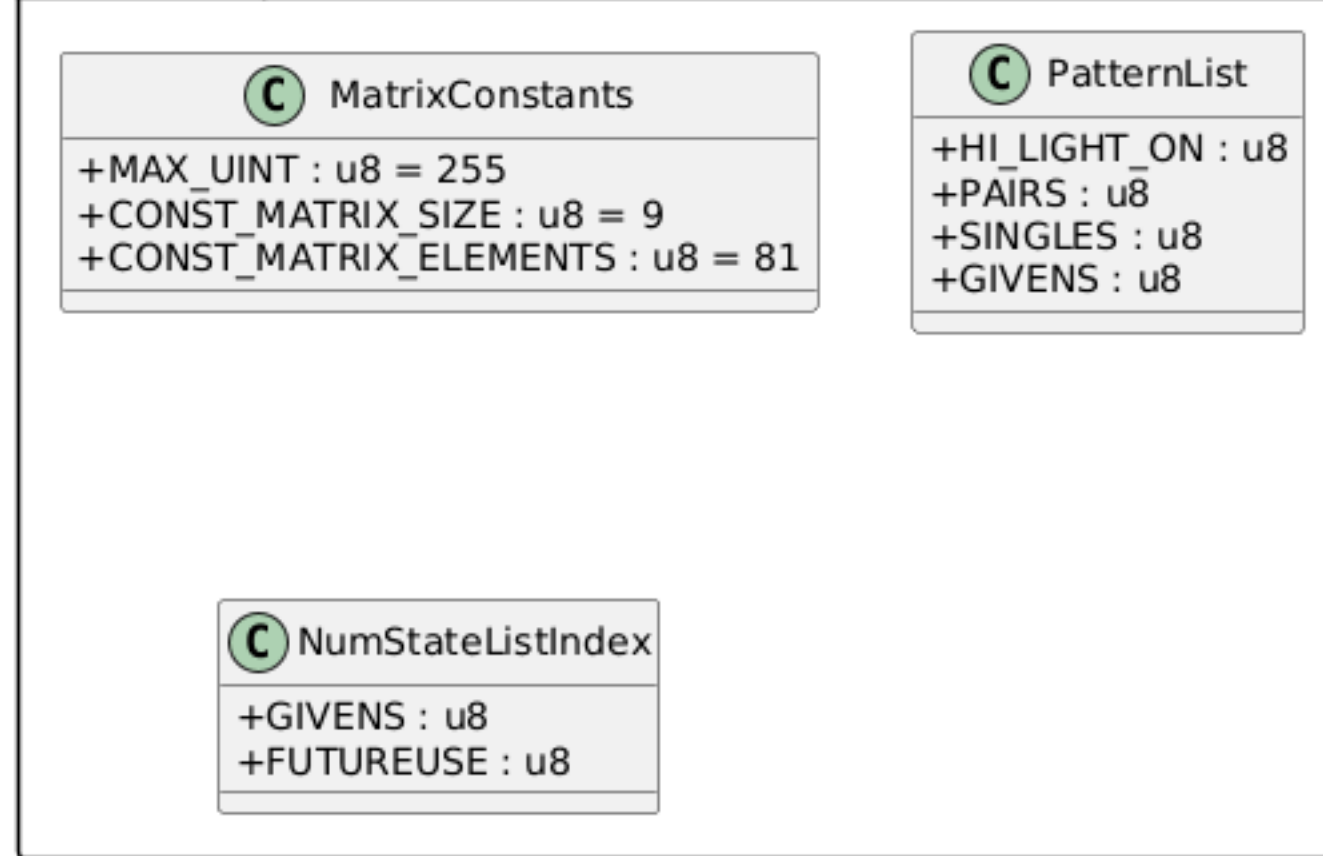
Runs solver analysis over entire matrix

FFI Data Model

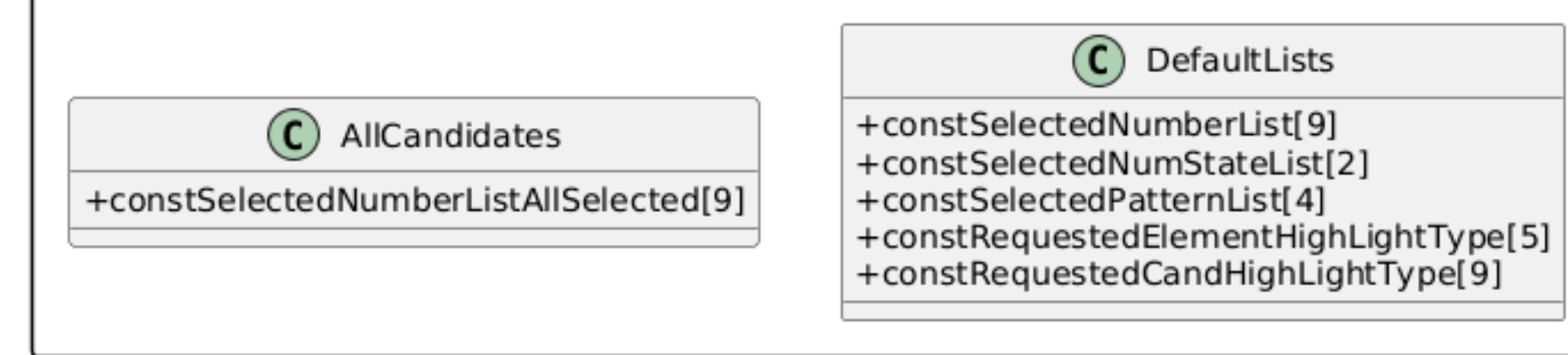


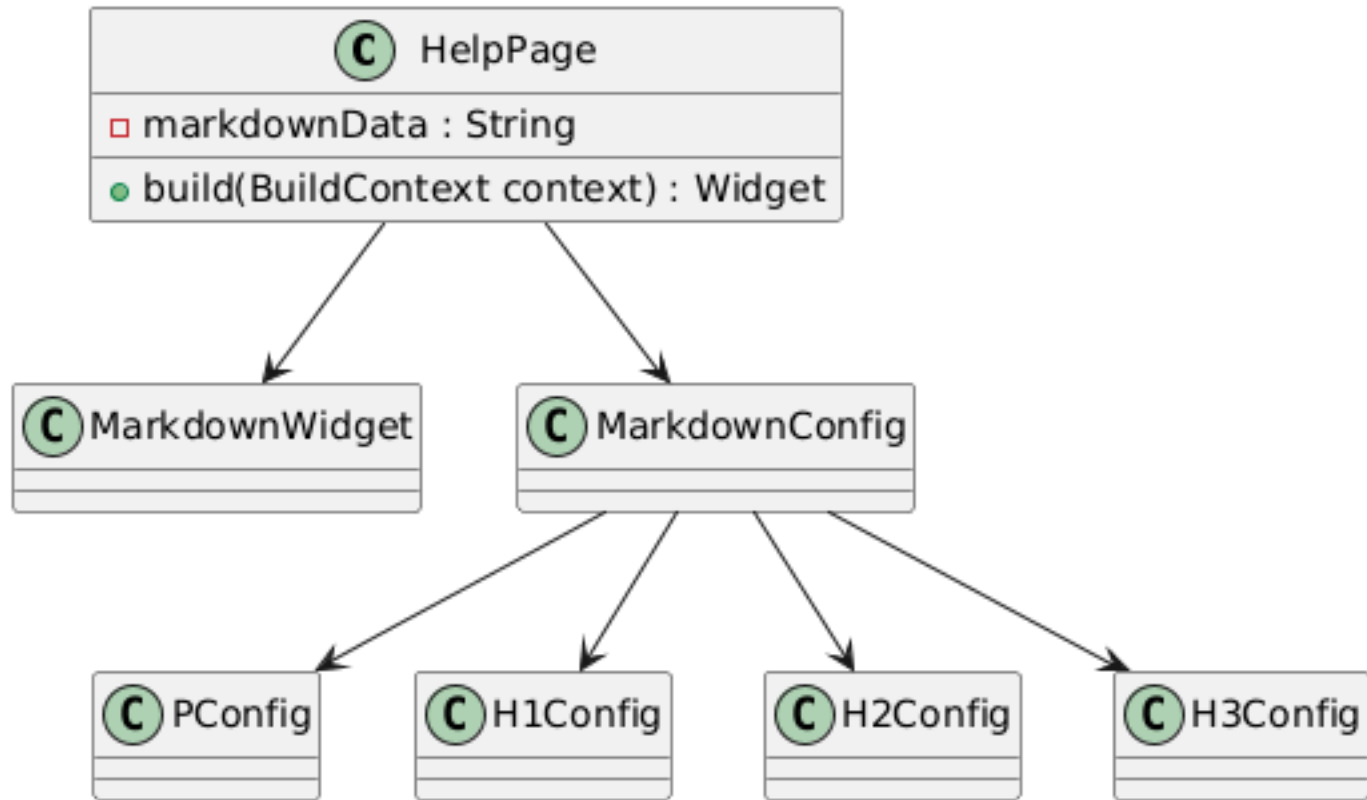
Represents a single Sudoku cell.
Used for FFI communication between Dart and Rust.
Total cells = 81
Matrix size = 9 x 9

Constants

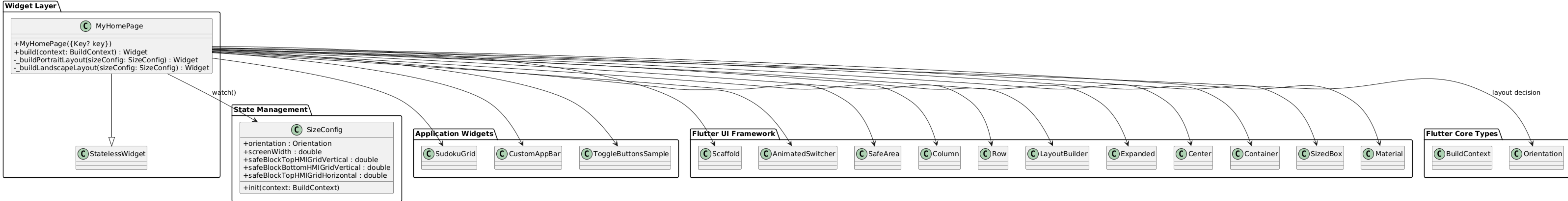


Default Data Arrays

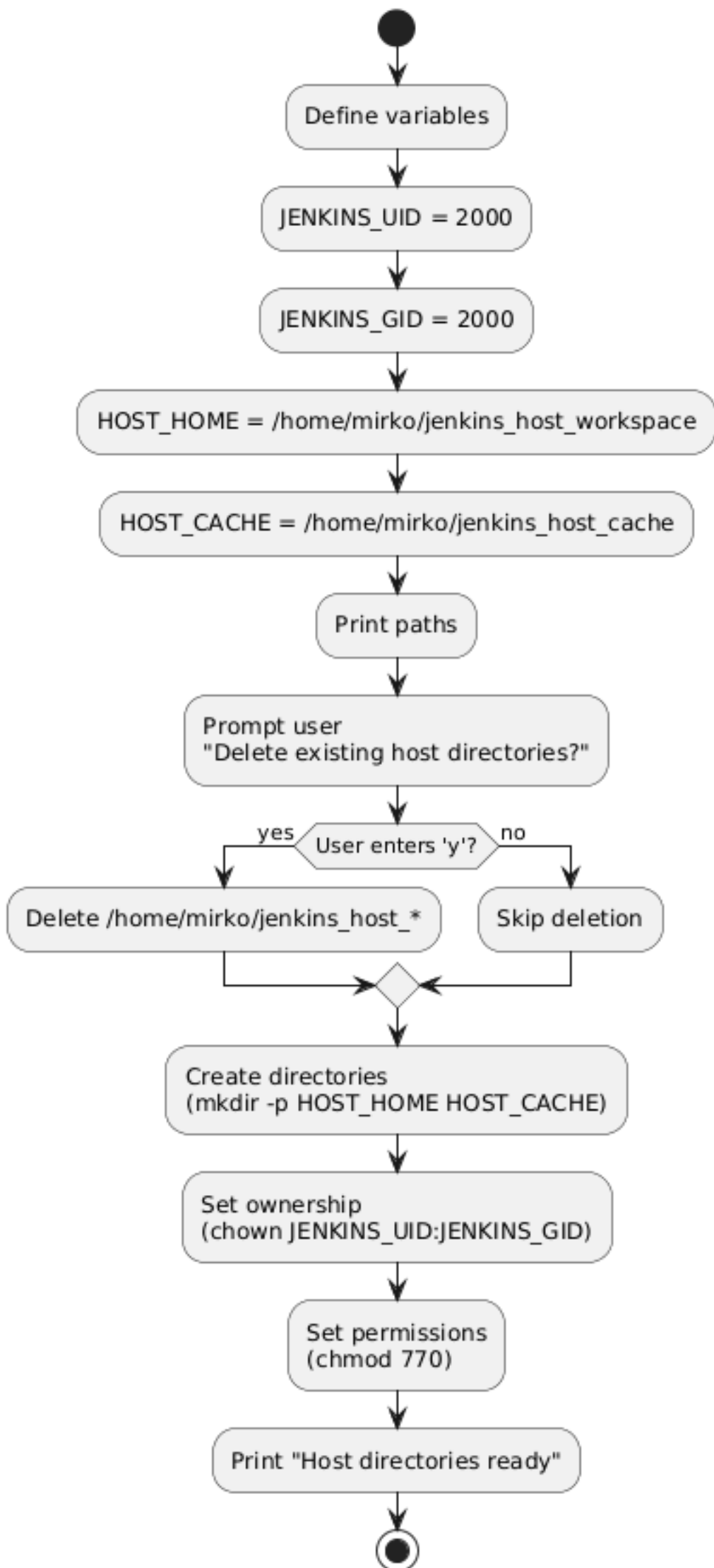




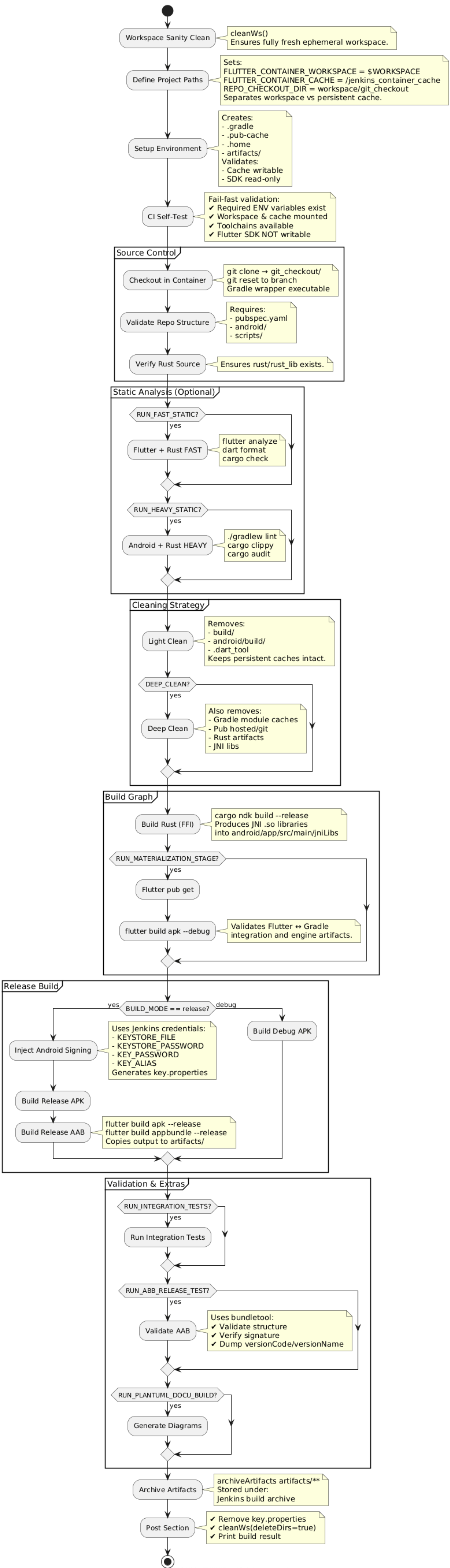
Flutter Widget Class Diagram - MyHomePage



Jenkins Host Directory Setup Script



Jenkins CI Pipeline - Flutter + Rust + Android




```
graph TD; VSCode[VSCode] --> FlutterCLI[FlutterCLI]; VSCode --> Gradle[Gradle]; VSCode --> Docker[Docker]; VSCode --> PlantUML[PlantUML];
```

VSCode

FlutterCLI

Gradle

Docker

PlantUML

Rust Sudoku Solver - Crate Module Structure

crate root (lib.rs)

lib.rs

Entry point of the Rust crate.

Defines internal modules:

- ffi
- process_data
- store_data

Re-exports the public API:
`pub use ffi::*;`

Dart / Flutter App

mod

FFI calls

Provides C-compatible API for Dart/Flutter via FFI.

Functions:

- create_matrix
- update_cell
- update_matrix
- erase_matrix
- set_all_candidates
- free_matrix

ffi.rs
(FFI interface)

mod

mod

uses

uses

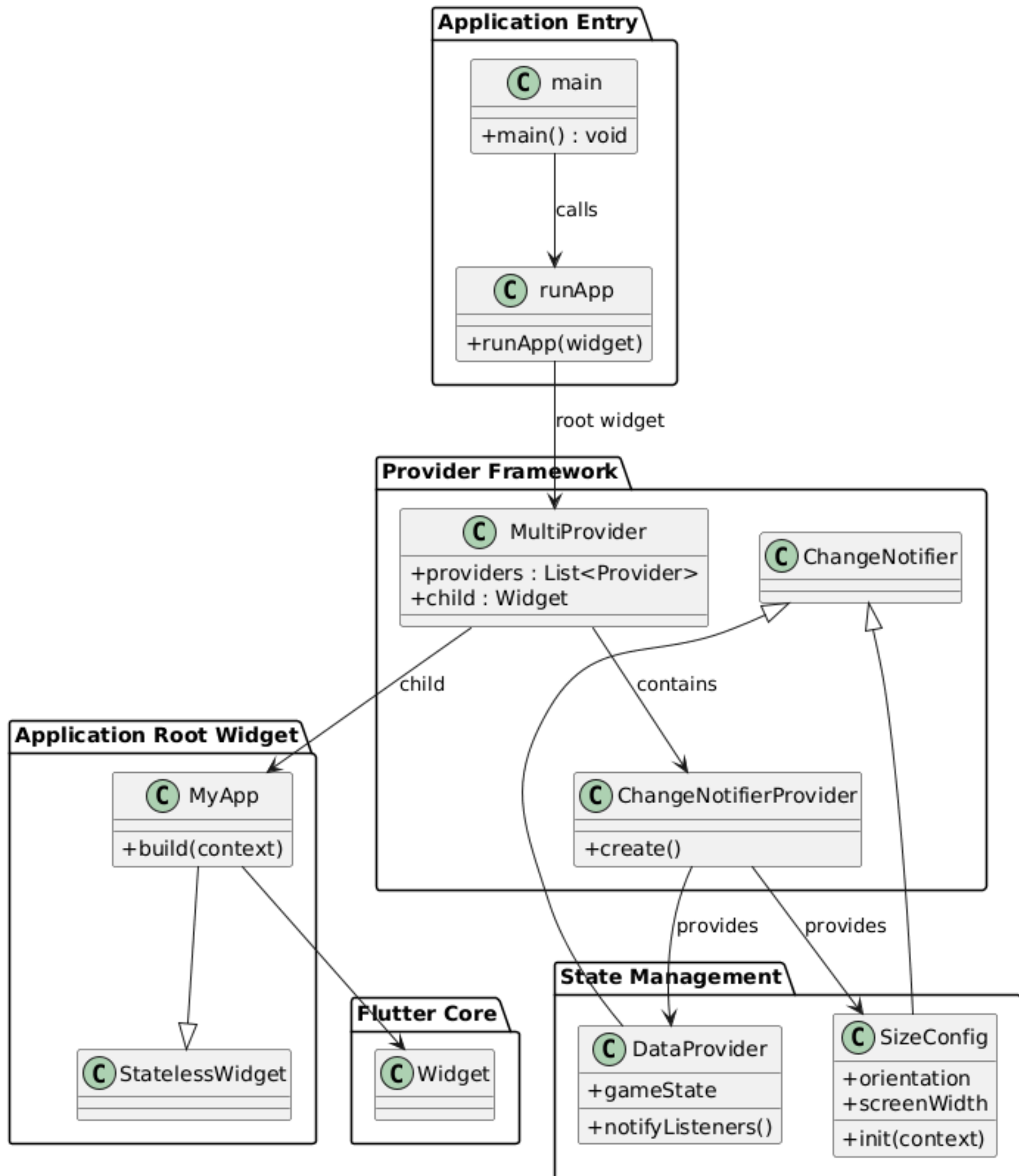
process_data.rs
(Sudoku solver logic)

store_data.rs
(JSON persistence)

Sudoku solving logic:
- check_one_element()
- check_all_elements()

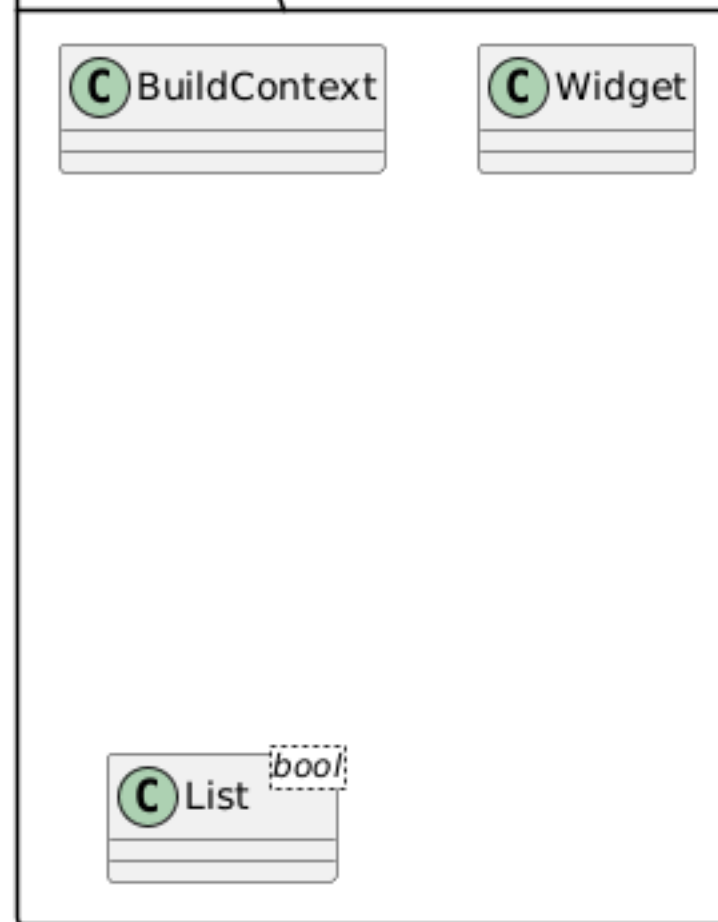
Handles persistence:
- load matrix
- save matrix
- JSON serialization

Flutter App Bootstrap & Provider Architecture

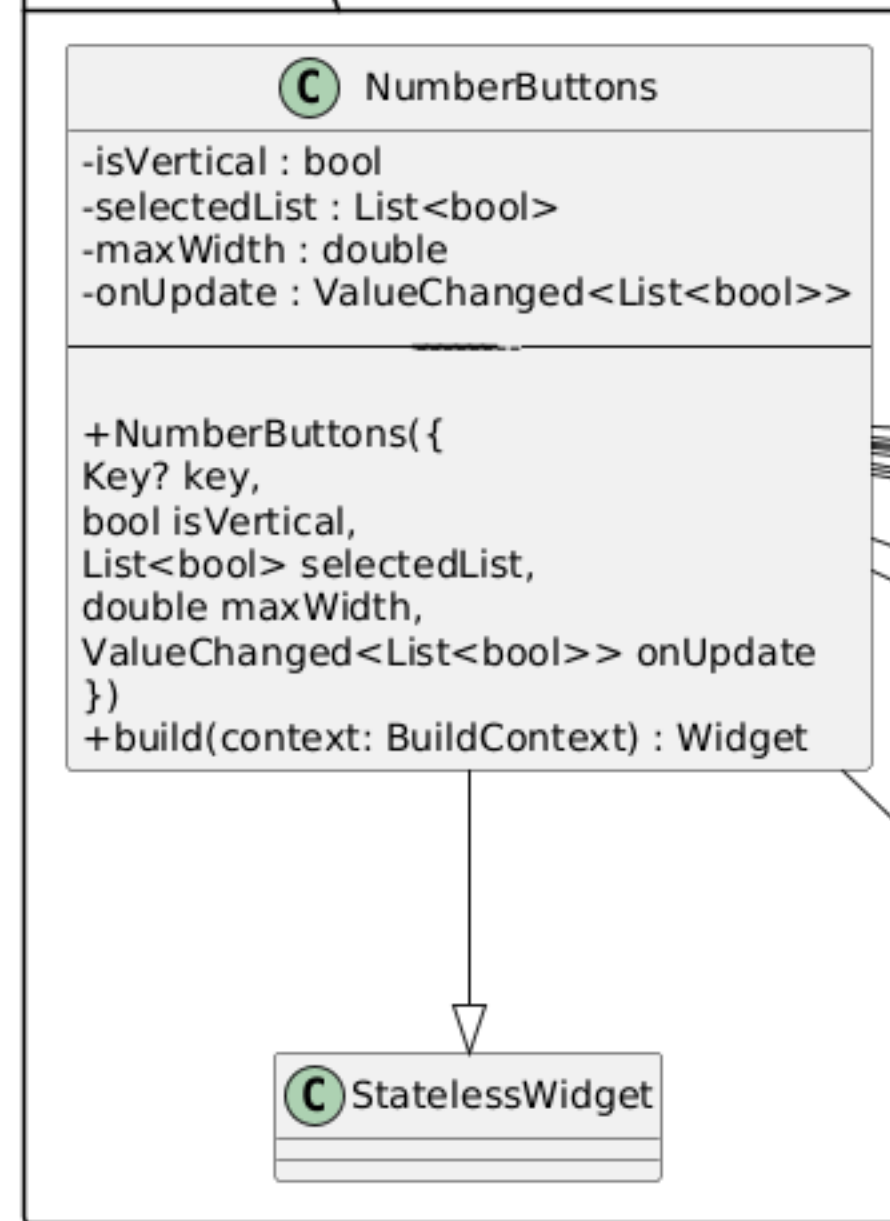


Flutter Widget Class Diagram - NumberButtons

Core Types



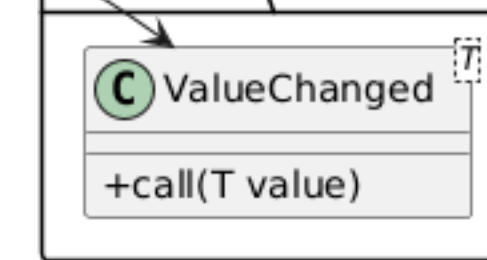
Widget Layer



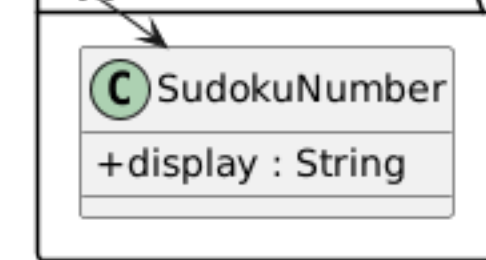
Flutter UI Framework



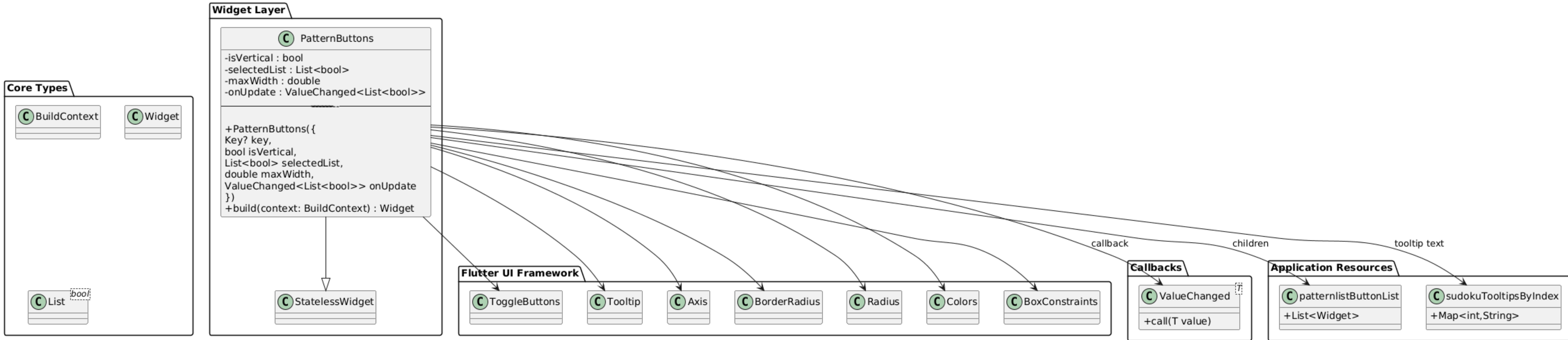
Callbacks



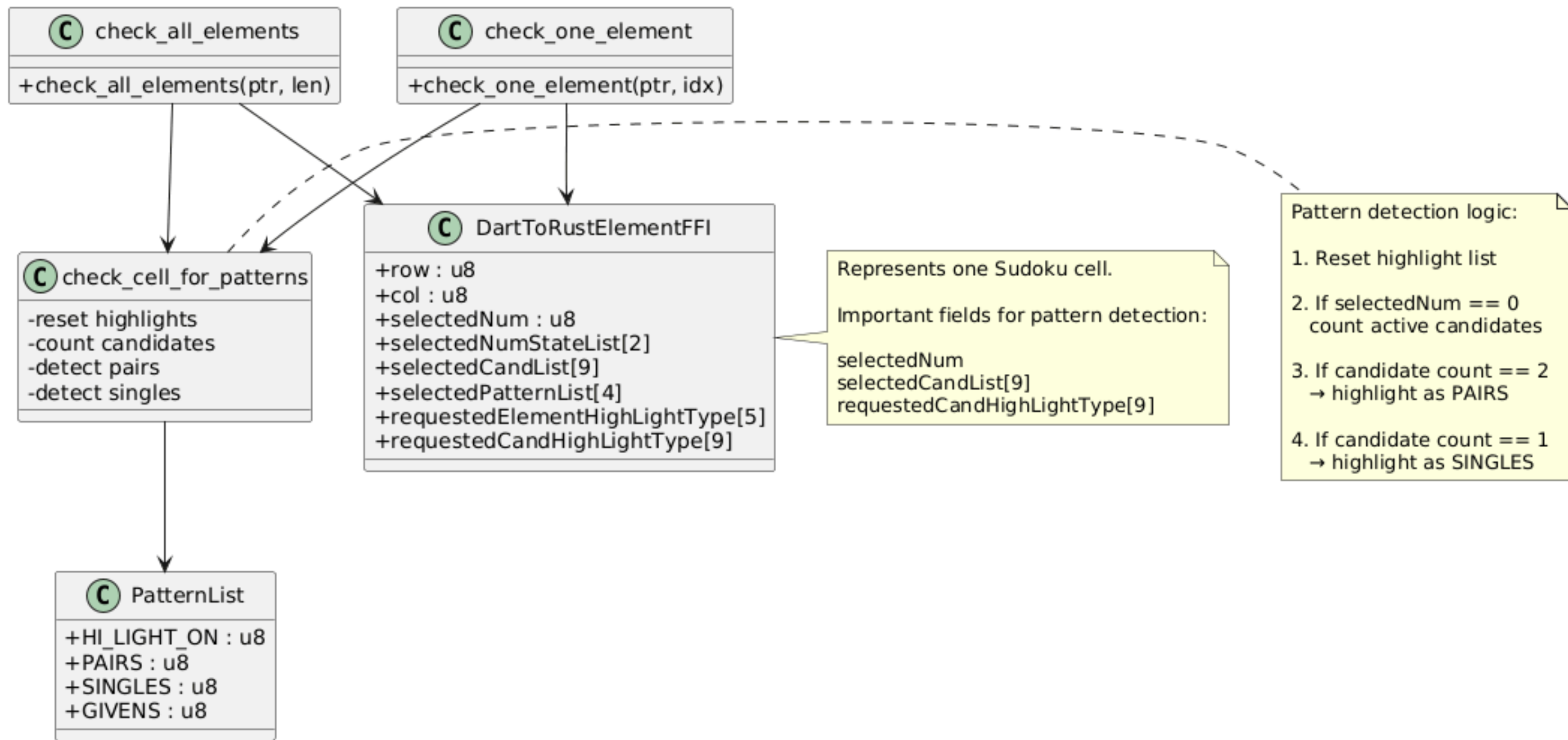
Application Domain



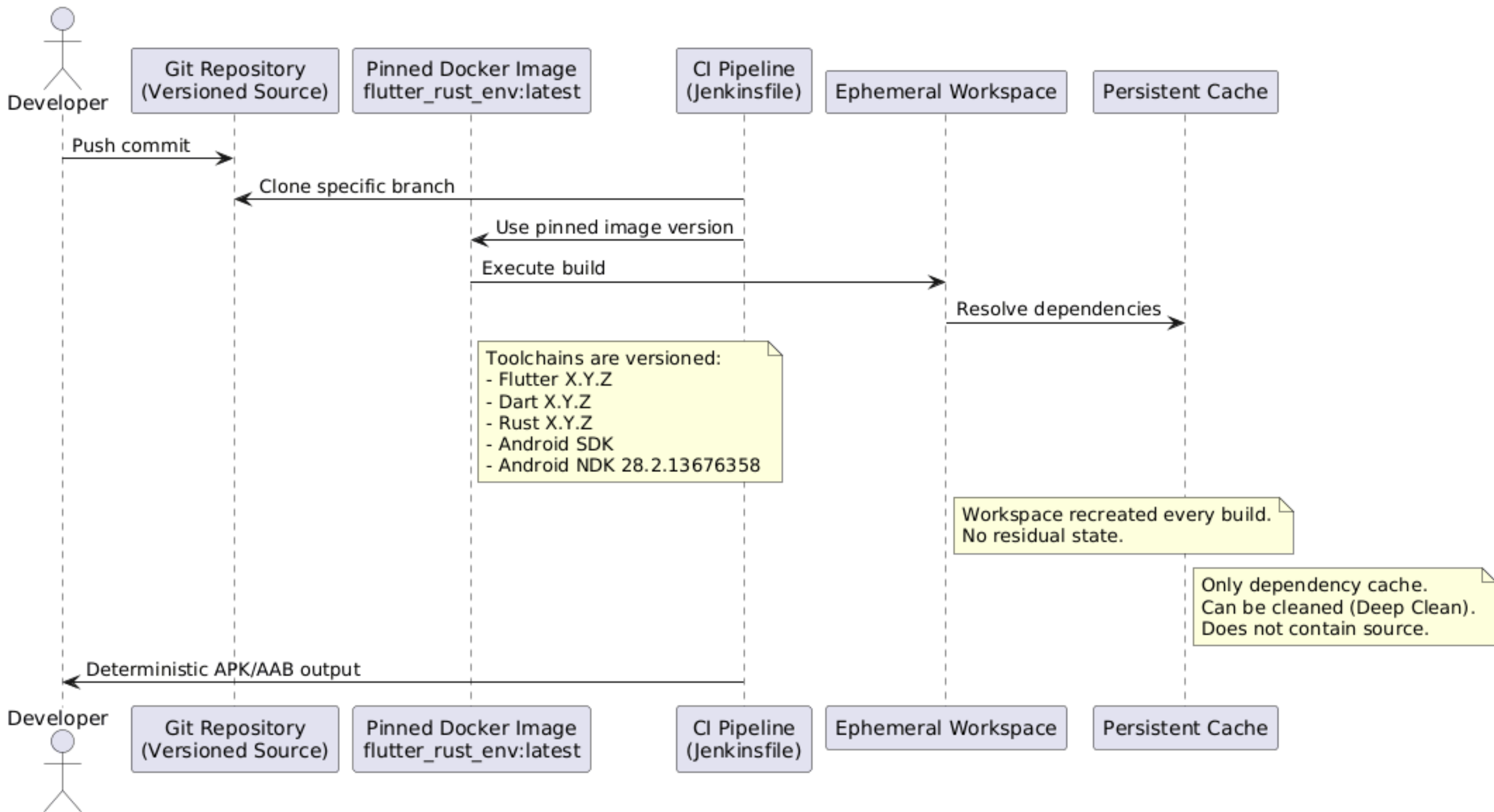
Flutter Widget Class Diagram - PatternButtons

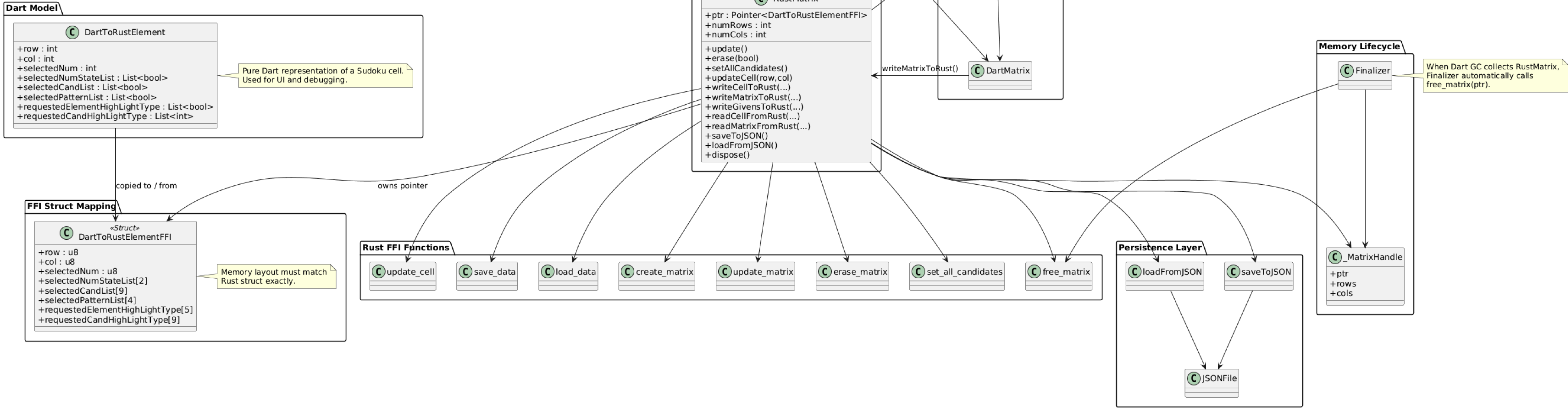


Sudoku Pattern Detection Engine (process_data.rs)

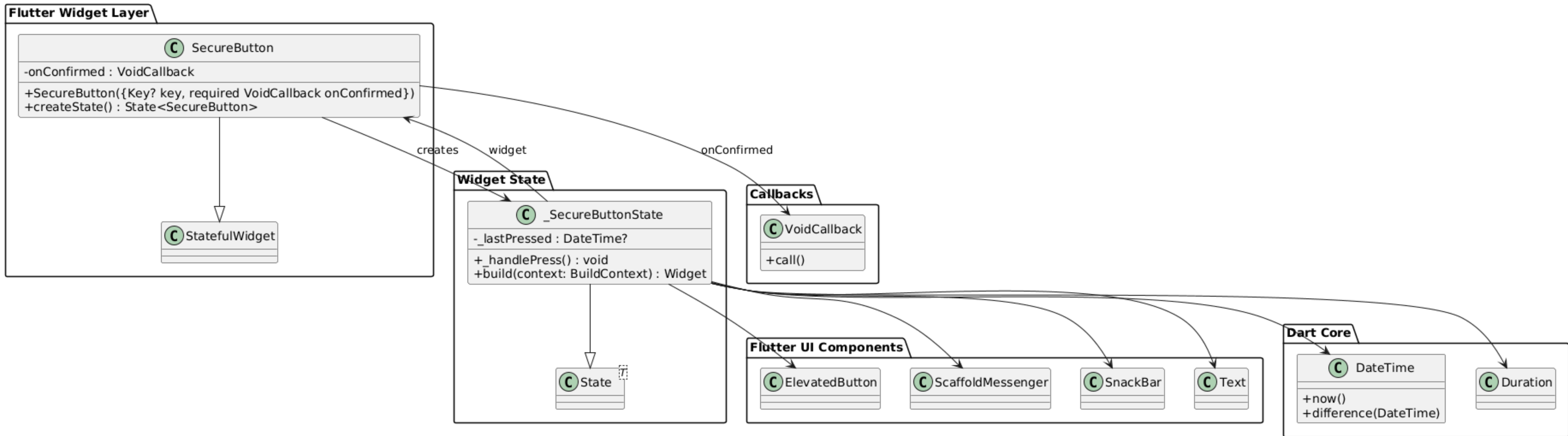


Deterministic Build Model

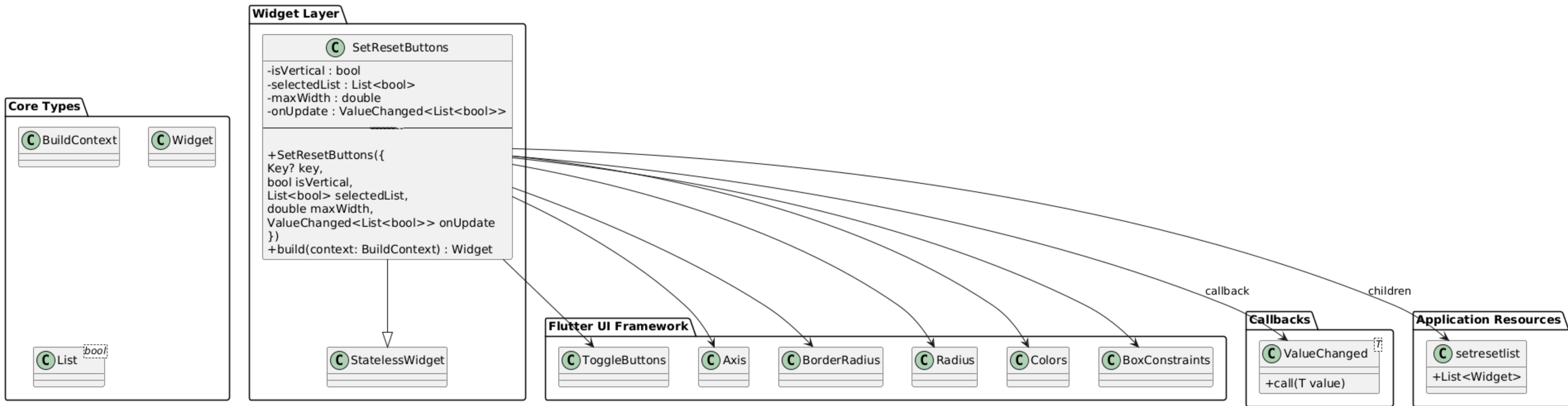




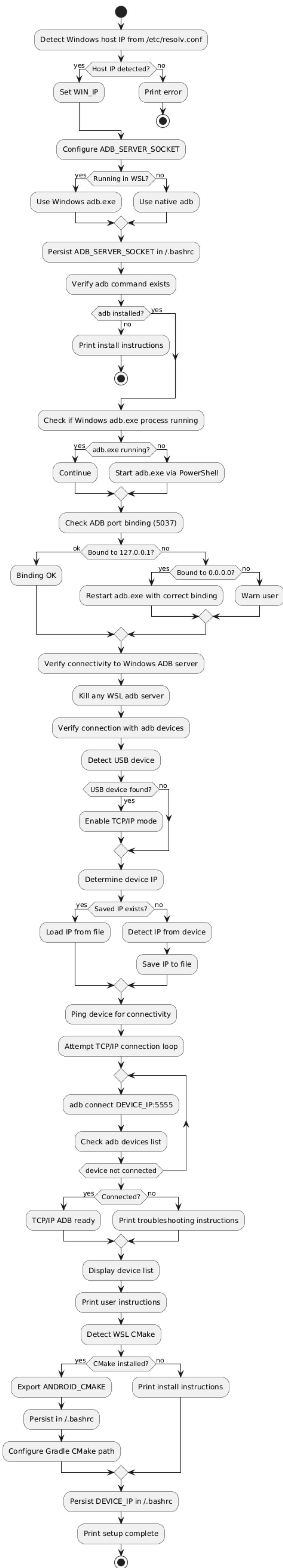
Flutter Widget Class Diagram - SecureButton



Flutter Widget Class Diagram - SetResetButtons



WSL ADB over TCP/IP Setup Workflow



Shared Types, Enums, Constants, and Utility Classes

C

GridPosition

int row

int col

GridPosition(int row, int col)

getRowColFromId(int id, int numColumns)
- Computes global Sudoku coordinates
- Based on 3×3 block subdivision

E

DataStatus

loading

ready

error

E

constIntNumList

ONE(1)

TWO(2)

THREE(3)

FOUR(4)

FIVE(5)

SIX(6)

SEVEN(7)

EIGHT(8)

NINE(9)

DEFAULT(255)

E

constIntCandList

ONE(1)

TWO(2)

THREE(3)

FOUR(4)

FIVE(5)

SIX(6)

SEVEN(7)

EIGHT(8)

NINE(9)

DEFAULT(255)

E

constTextNumList

ONE("1")

TWO("2")

THREE("3")

FOUR("4")

FIVE("5")

SIX("6")

SEVEN("7")

EIGHT("8")

NINE("9")

DEFAULT("255")

E

SudokuNumber

one(1)

two(2)

three(3)

four(4)

five(5)

six(6)

seven(7)

eight(8)

nine(9)

E

constIntPatternList

hiLightOn(0)

pairs(1)

singles(2)

givens(3)

DEFAULT(255)

«typedef»

C

SelectedNumberList

List<bool>

«typedef»

C

SelectedNumStateList

List<bool>

«typedef»

C

SelectedCandList

List<bool>

«typedef»

C

SelectedSetResetList

List<bool>

«typedef»

C

SelectedPatternList

List<bool>

«typedef»

C

RequestedElementHighLightType

List<bool>

«typedef»

C

RequestedCandHighLightType

List<int>

«typedef»

C

SelectedUndoIconList

List<bool>

«typedef»

C

SelectedAddRemoveList

List<bool>

C

PatternList

hiLightOn = 0

pairs = 1

singles = 2

givens = 3

C

undoiconlistIndex

undo = 0

redo = 1

C

SelectedNumStateListIndex

Givens = 0

FutureUse = 1

C

addRemoveListIndex

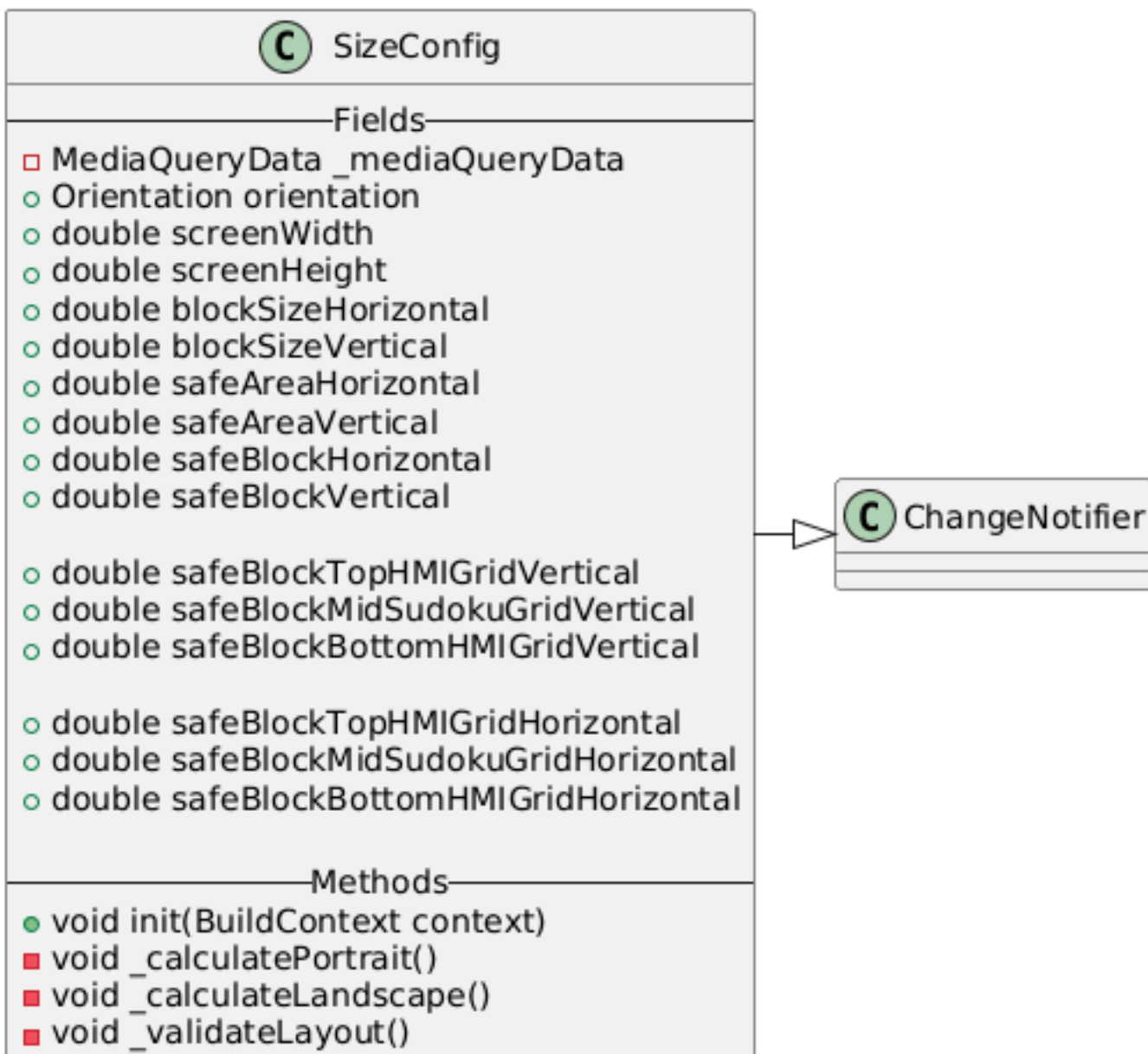
saveGivens = 0

eraseAll = 1

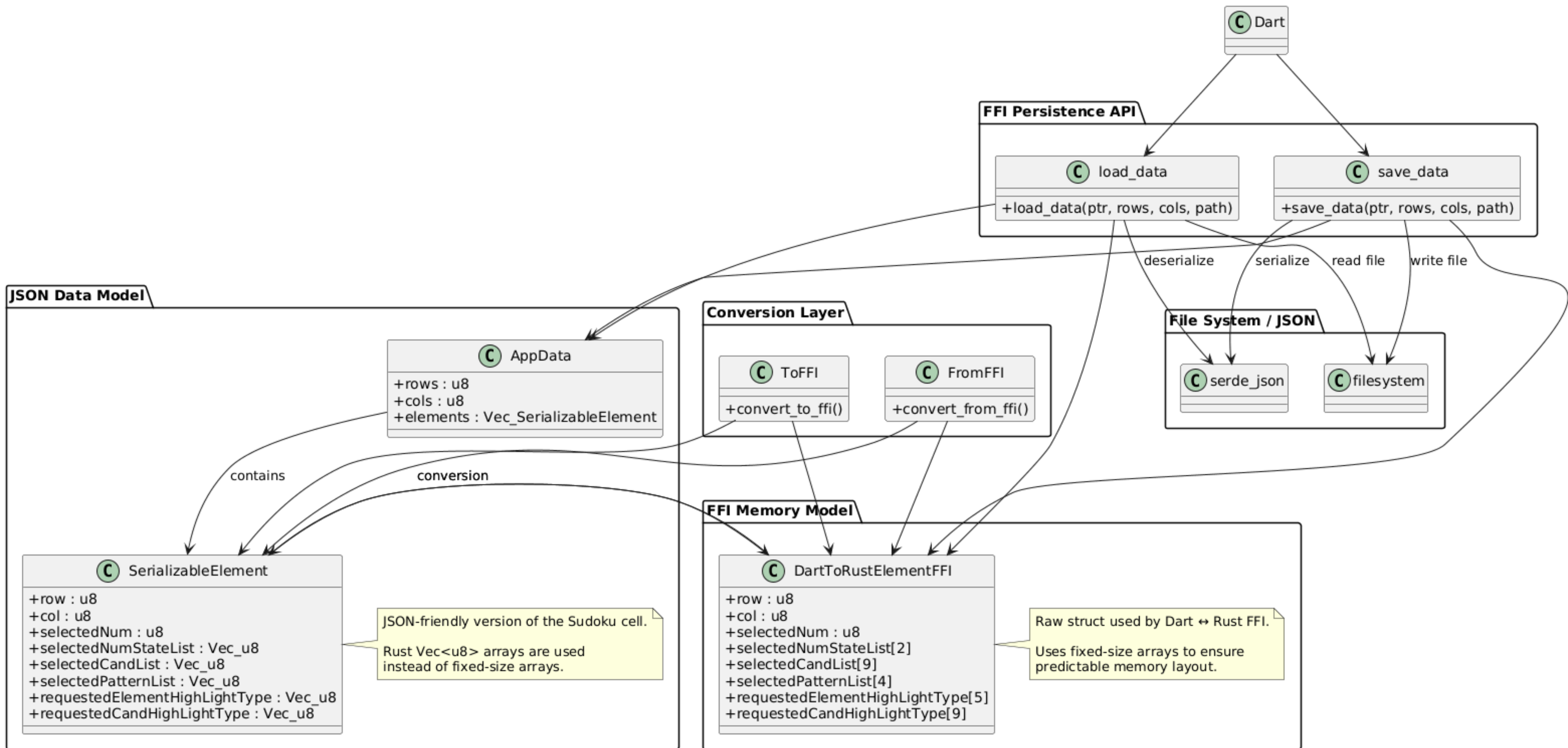
resetToGivens = 2

selectAllCand = 3

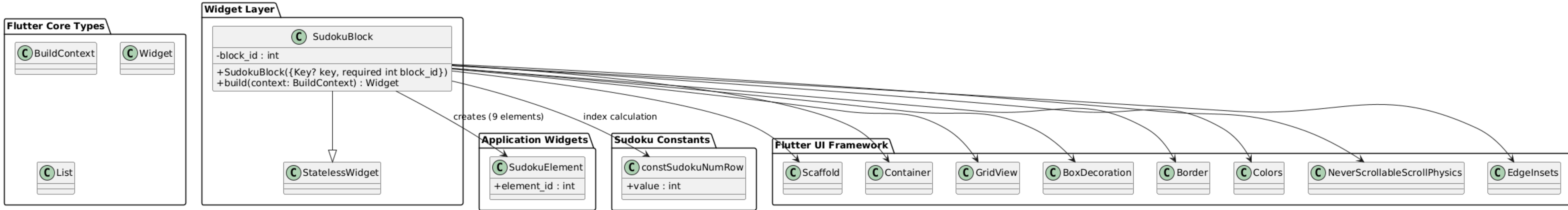
SizeConfig Class Diagram

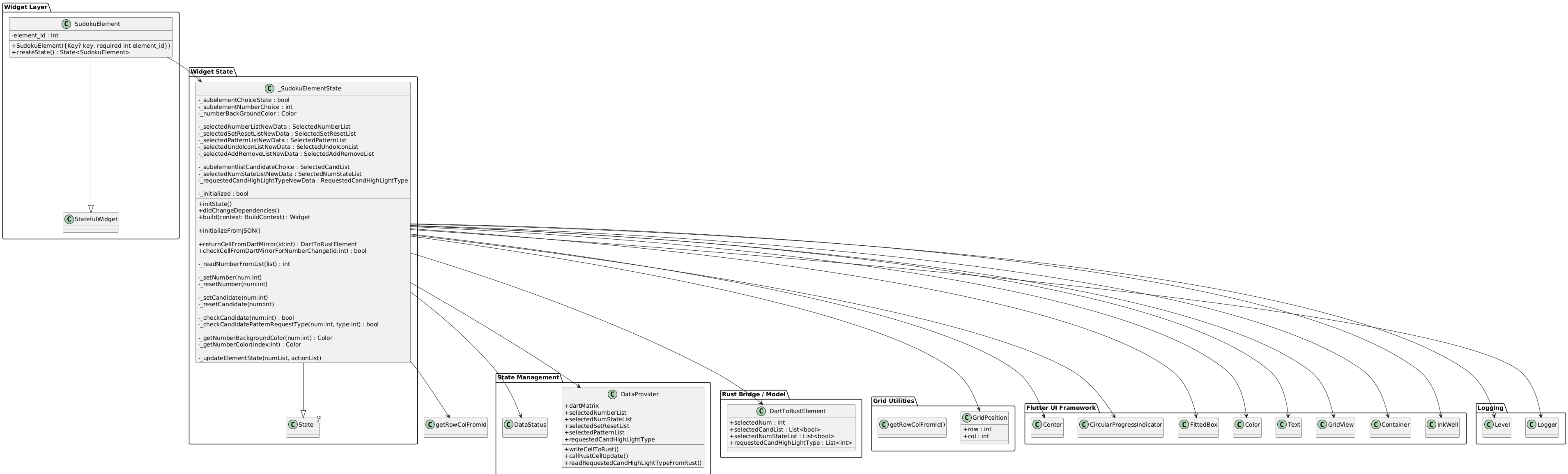


Sudoku Data Persistence (store_data.rs)

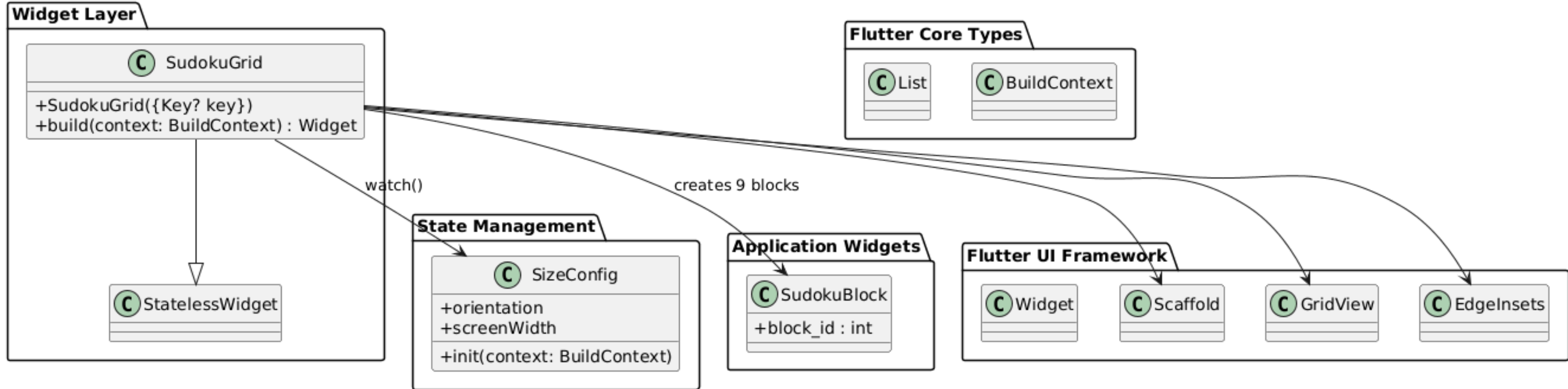


Flutter Widget Class Diagram - SudokuBlock





Flutter Widget Class Diagram - SudokuGrid



Initializes global logging for debugging and diagnostics.

Debug build:
Logger.root.level = Level.ALL

Release build:
Logger.root.level = Level.SEVERE

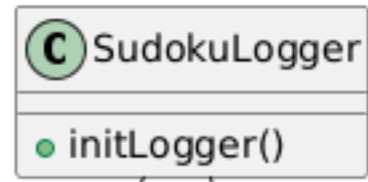
Example usage:
log.info("Matrix loaded")
log.warning("Unexpected state")
log.severe("Rust call failed")

Other options :

- FINEST
- FINER
- FINE
- CONFIG
- INFO
- WARNING
- SEVERE
- SHOUT

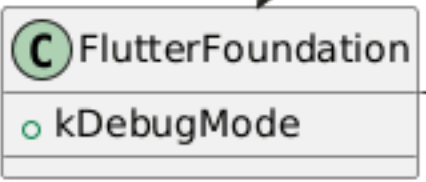
Log visibility:

info -> Debug: shown, Release: hidden
warning -> Debug: shown, Release: hidden
severe -> Debug: shown, Release: shown

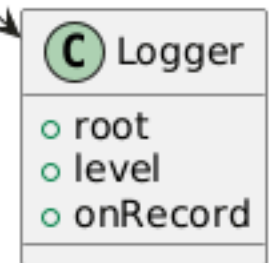


reads

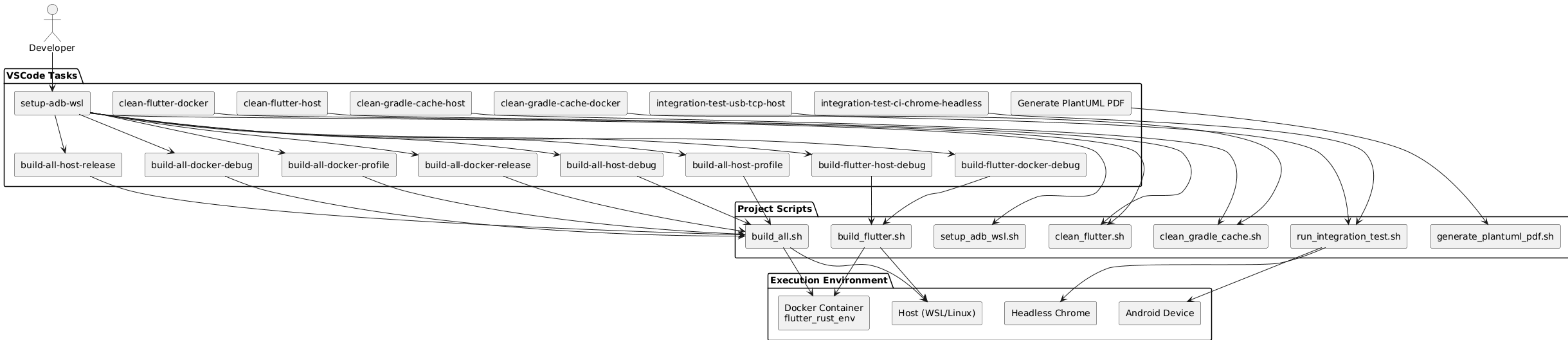
configures



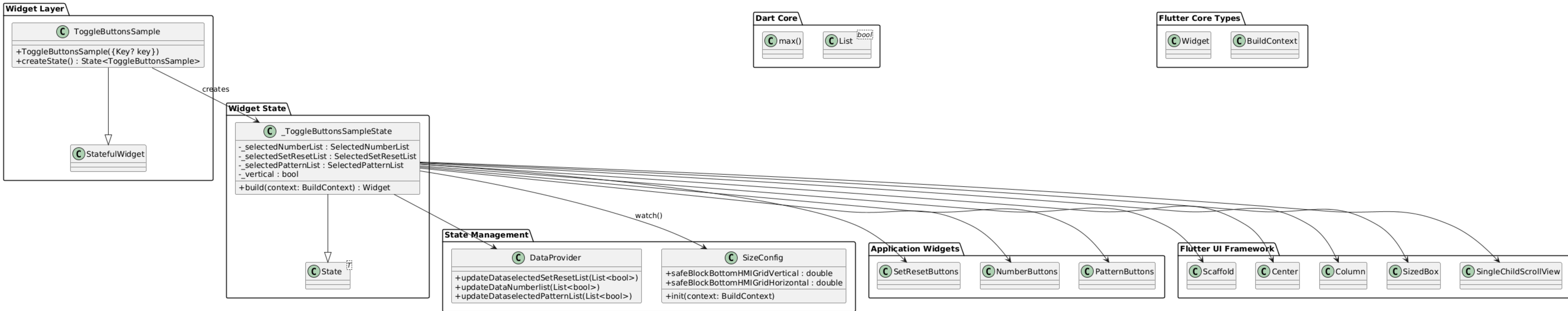
Provided by Flutter.
kDebugMode determines whether the application runs in debug or release mode.



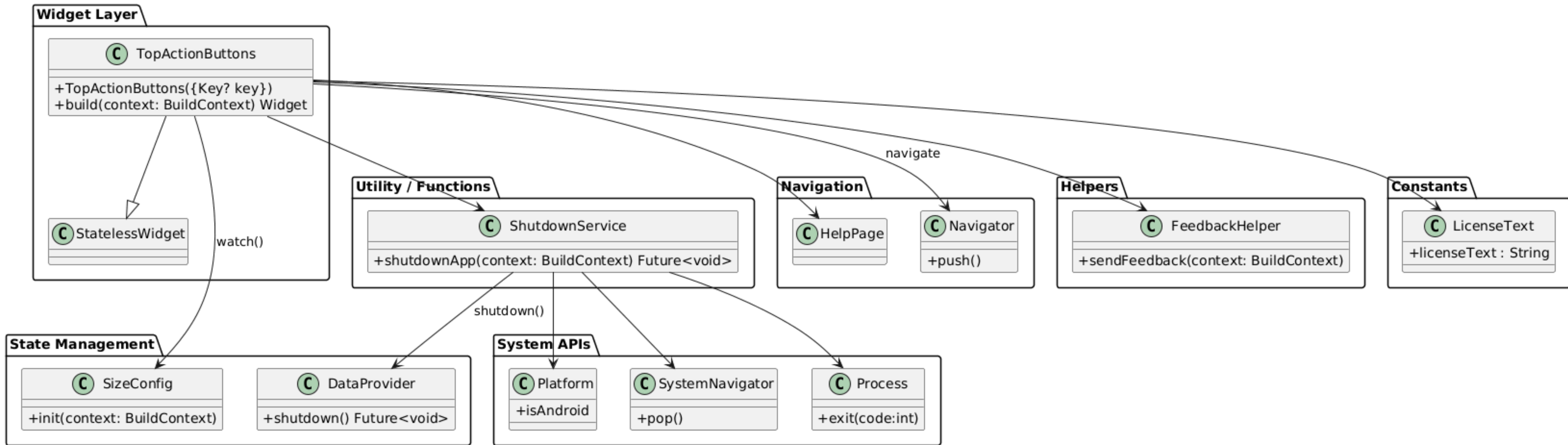
VSCode Tasks - Flutter + Rust Build Pipeline



Flutter Widget Class Diagram - ToggleButtonsSample

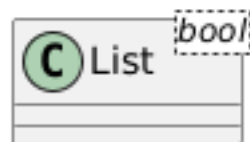


Flutter Widget Class Diagram - TopActionButtons



Flutter Widget Class Diagram - UndoIconsButtons

Core Types



Widget Layer

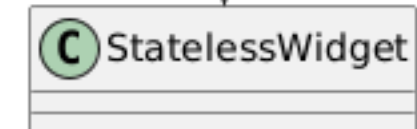
UndoIconsButtons

-isVertical : bool
-selectedList : List<bool>
-onUpdate : ValueChanged<List<bool>>

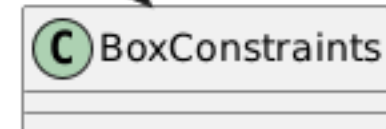
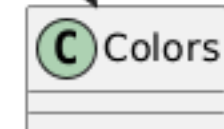
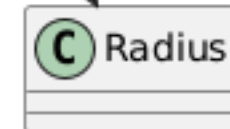
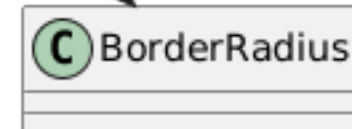
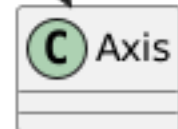
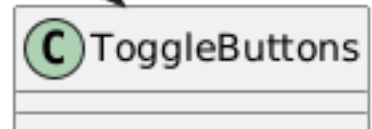
Key? key,
bool isVertical,
List<bool> selectedList,
ValueChanged<List<bool>> onUpdate

+UndoIconsButtons({
})

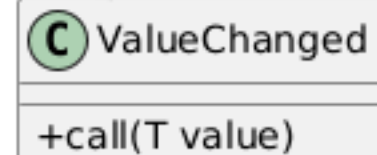
+build(context: BuildContext) : Widget



Flutter UI Framework



Callbacks



Application Resources



C FeedbackHelper

• sendFeedback(BuildContext)

Launches the email client
to send user feedback.

C Uri

C url_launcher

Workspace vs Cache Security Model

